Sustainable NN & DL: Frontier Intelligence With Less Compute

Danna Gurari
University of Colorado Boulder
Spring 2026

CC BY NC SA

https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last lecture:
  - Guest talk from Eric Hersh, Principal Water Resources Engineer at Stantec, about data centers and mitigating their adverse impacts to the environment and society
- Assignments (Canvas):
  - Final project outline due in 1 week
- Questions?

Today’s Topics

- The shock: frontier AI without the frontier budget
- Three technical levers for efficient intelligence
- The bigger story: a new scaling paradigm

Today’s Topics

- The shock: frontier AI without the frontier budget
- Three technical levers for efficient intelligence
- The bigger story: a new scaling paradigm

Frontier Models: Foundation Models That Are the Most Advanced at a Given Time

- What comes to mind?
  - Open AI's GPT-5 series (5.2 thinking and 5.3 codex)
  - Claude 4 series (Opus, Sonnet, and Haiku)
  - Gemini 3.1

Jan 2025: The Release of DeepSeek-R1 Changed the World's Expectations for AI…



https://www.youtube.com/watch?v=AK0o4LaEvV8

- Within 2 weeks of launch, 30M+ daily active users;
- Within 20 days, 100M+ users
- How much do you think was spent on advertising?

Read time: 9 min

OpenAI's o1 vs. DeepSeek-R1: Which Should You Choose?

February 21, 2025

https://www.knime.com/blog/openai-o1-vs-deepseek-r1

DeepSeek R1 Basically Replaces GPT O1 - for free

Rivi Richards

Updated:2026-03-08 5mins

https://www.reddit.com/r/LocalLLaMA/comments/1ikyq45/deepseek_gained_over_100_millions_users_in_20_days/

Jan 2025: The Release of DeepSeek-R1 Changed the World's Expectations for AI…



https://www.youtube.com/watch?v=AK0o4LaEvV8

Released January 20, 2025

AI EFFECT

Nvidia sheds almost $600 billion in market cap, biggest one-day loss in U.S. history

PUBLISHED MON JAN 27 2025•4:08 PM EST UPDATED MON, JAN 27 2025•5:26 PM EST

Samantha Subin
@SAMANTHA_SUBIN

SHARE

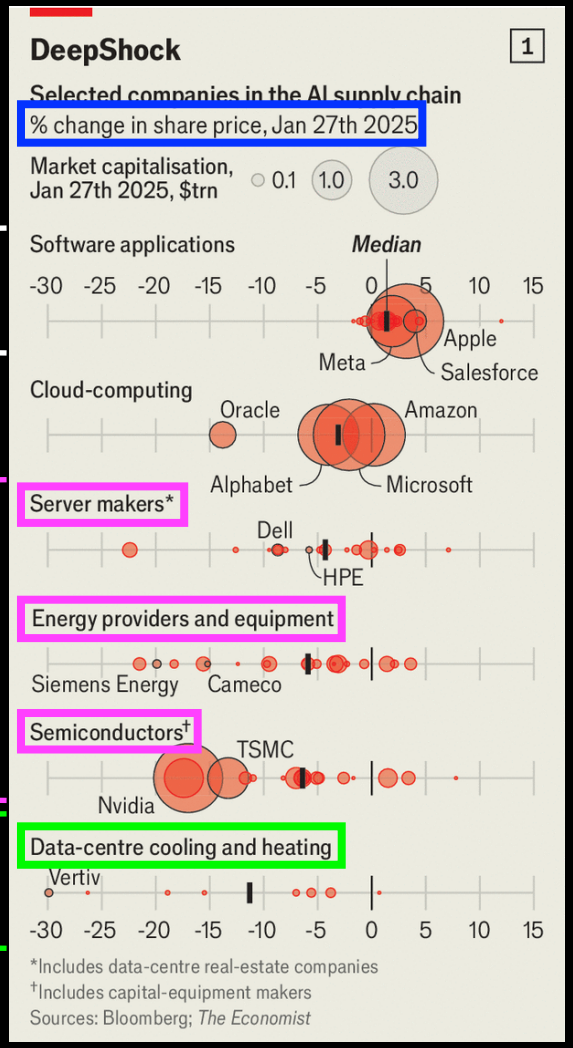
"The sell-off, which hit much of the U.S. tech sector, was sparked by concerns about increased competition from Chinese AI lab DeepSeek."

Jan 2025: The Release of DeepSeek-R1 Changed the World's Expectations for AI…

Why would SW makers see profits?

"Other businesses in the AI supply chain… suffered a similar fate. The cause of investors' panic was DeepSeek… Innovative techniques have allowed the firm to train AI models that perform about as well as the most sophisticated Western ones with only a fraction of the computing power—and therefore a fraction of the cost."

Why would energy firms see losses?



https://www.economist.com/business/2025/01/27/deepseek-sends-a-shockwave-through-markets

DeepSeek R1 vs OpenAI O1

Dimension	DeepSeek R1	OpenAI O1
License	Open source (MIT License)	Proprietary
Model Transparency	Full weights and architecture visible	Black box implementation
Development Cost	$5.6M reported training cost	Estimated hundreds of millions
Usage Pricing	$0.14/1M input, $2.19/1M output tokens	$15/1M input, $60/1M output tokens
Deployment Options	Self-host, on-premise, cloud	API-only, managed service
Context Length	128K tokens	128K tokens
Parameters	671B (mixture-of-experts)	Undisclosed dense architecture
Reasoning Approach	Chain-of-thought + reinforcement learning	Proprietary reasoning optimization
Data Sovereignty	Complete control	Data processed by OpenAI
Customization	Full fine-tuning capability	Limited to API parameters
Enterprise Support	Community-driven	Dedicated enterprise support
Integration Ecosystem	Growing open-source tools	Mature SDK and plugin ecosystem
Compliance Controls	Self-managed audit trails	OpenAI's compliance framework
Scaling Responsibility	Your infrastructure management	Automatic OpenAI scaling

What techniques enable frontier AI with low training and inference cost?

https://galileo.ai/blog/deepseek-r1-vs-openai-o1-comparison

Today’s Topics

- The shock: frontier AI without the frontier budget
- Three technical levers for efficient intelligence
- The bigger story: a new scaling paradigm

(December 2024)

DeepSeek-V3 Technical Report

DeepSeek-AI

(January 2025)

deepseek

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI



DeepSeek-V3

Mixture-of-experts + more architecture and HW tricks



DeepSeek-R1

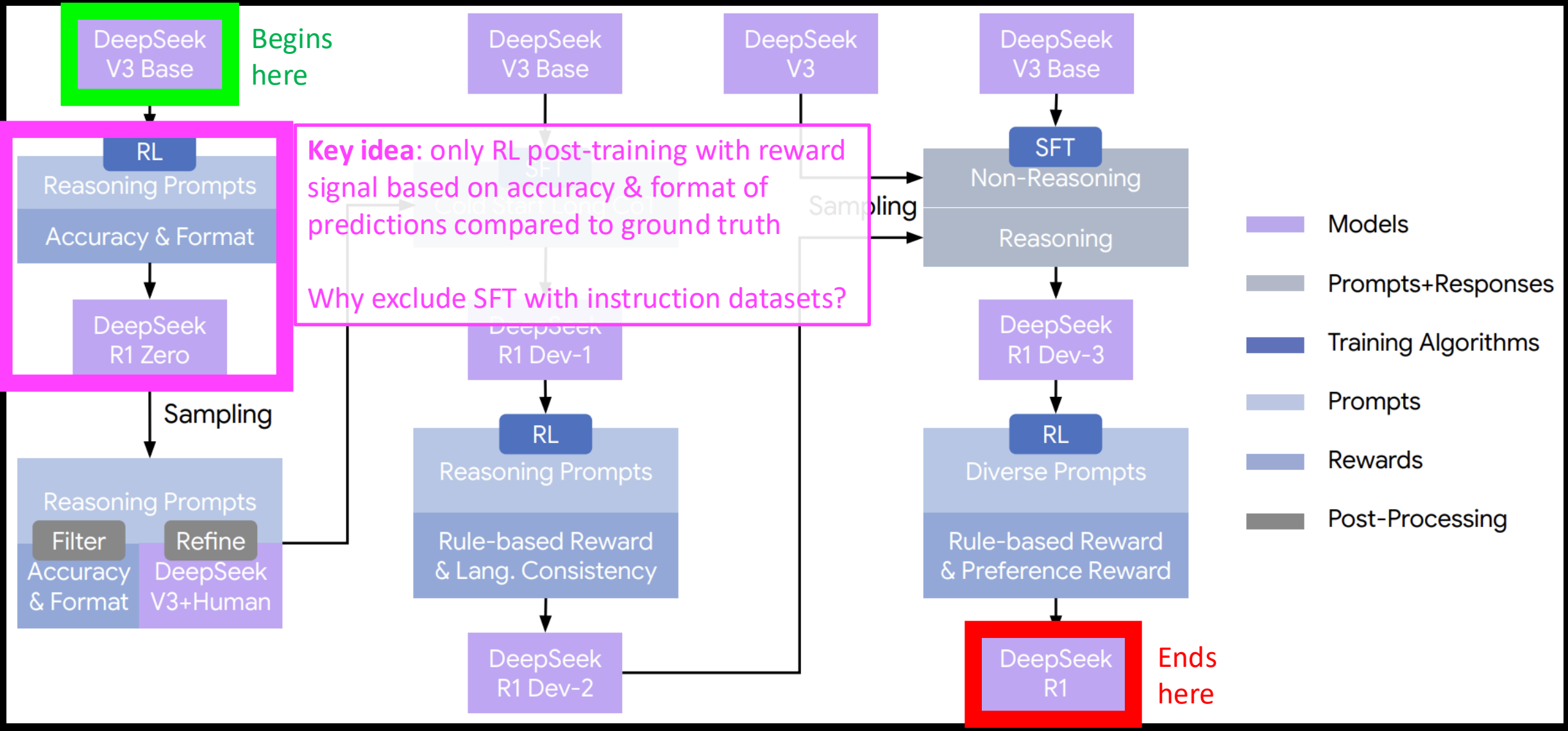
Frontier-level reasoning with small training cost



DeepSeek-R1-Distill Series

Smaller variants

https://www.etsy.com/listing/1172948287/vintage-toys-collectible-miniature



Leads to frontier-level reasoning with relatively small training cost

DeepSeek-R1 paper. 2025

Prompt: Requires Reasoning and Answer

A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within `<think>...</think>` and `<answer>...</answer>` tags, respectively, i.e., `<think>` reasoning process here `</think>` `<answer>` answer here `</answer>`. User: prompt. Assistant:

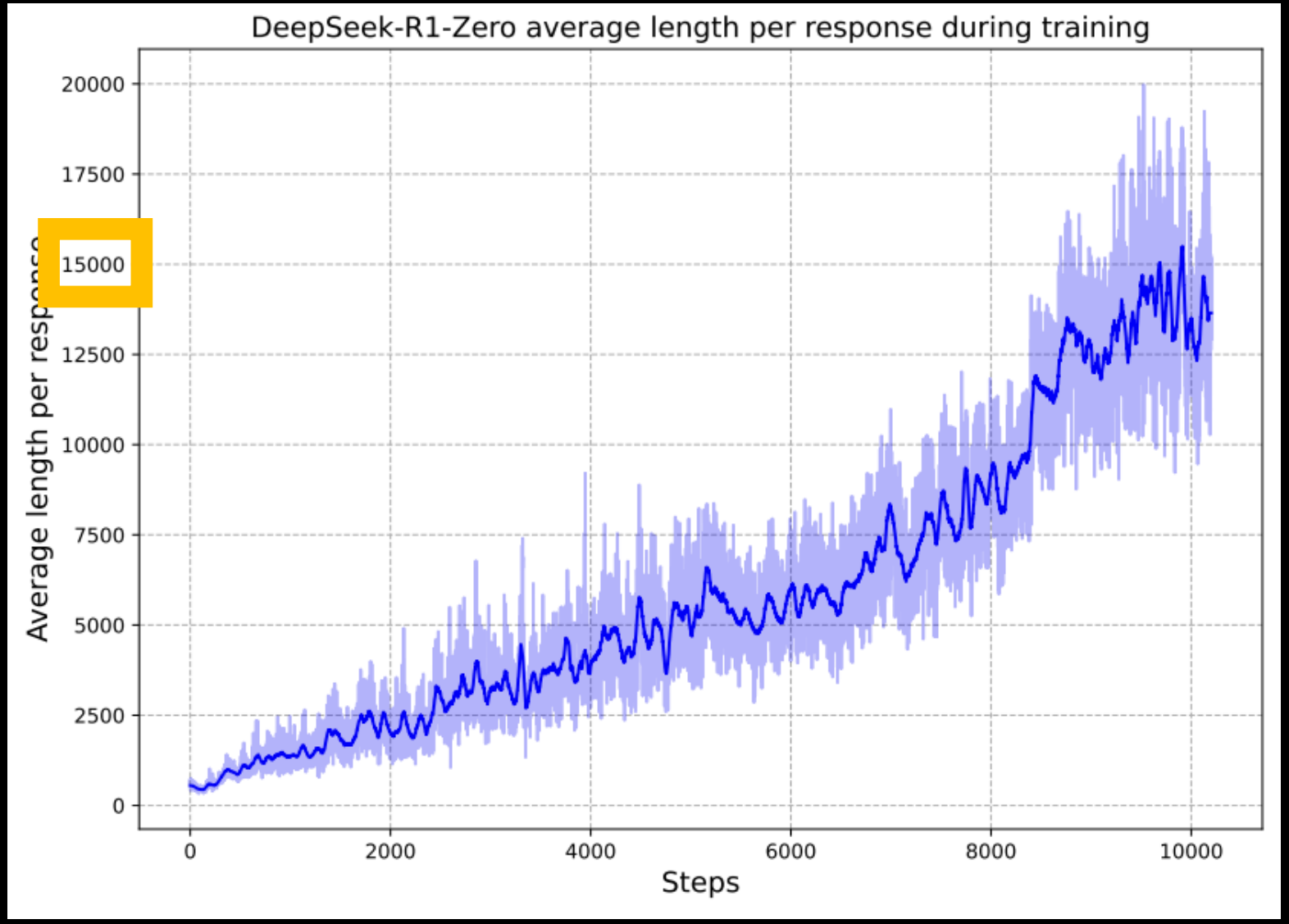
Data sources in Chinese and English:

Data Type	# Prompts	Question Type	Output Type
Math	26K	Quantitative Reasoning	Number/Expression/Equation
Code	17K	Algorithm and Bug Fixing	Code Solution
STEM	22K	Multi-Choice	Option
Logic	15K	Choice/Quantitative Reasoning	Option/Number
General	66K	Helpfulness/Harmlessness	Ranked Responses

DeepSeek-R1 paper. 2025

Analysis of RL Training

≈ 20 single spaced pages
if ~11,000 words with
~250 words per page



What is the trend for response length as training time grows?
Up to how many tokens are predicted?

DeepSeek-R1 paper. 2025

Analysis of RL Training

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: `<think>`

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$

$$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$$

$\ldots$

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$

We started with the equation:

$$\sqrt{a - \sqrt{a + x}} = x$$

First, let's square both sides:

$$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$$

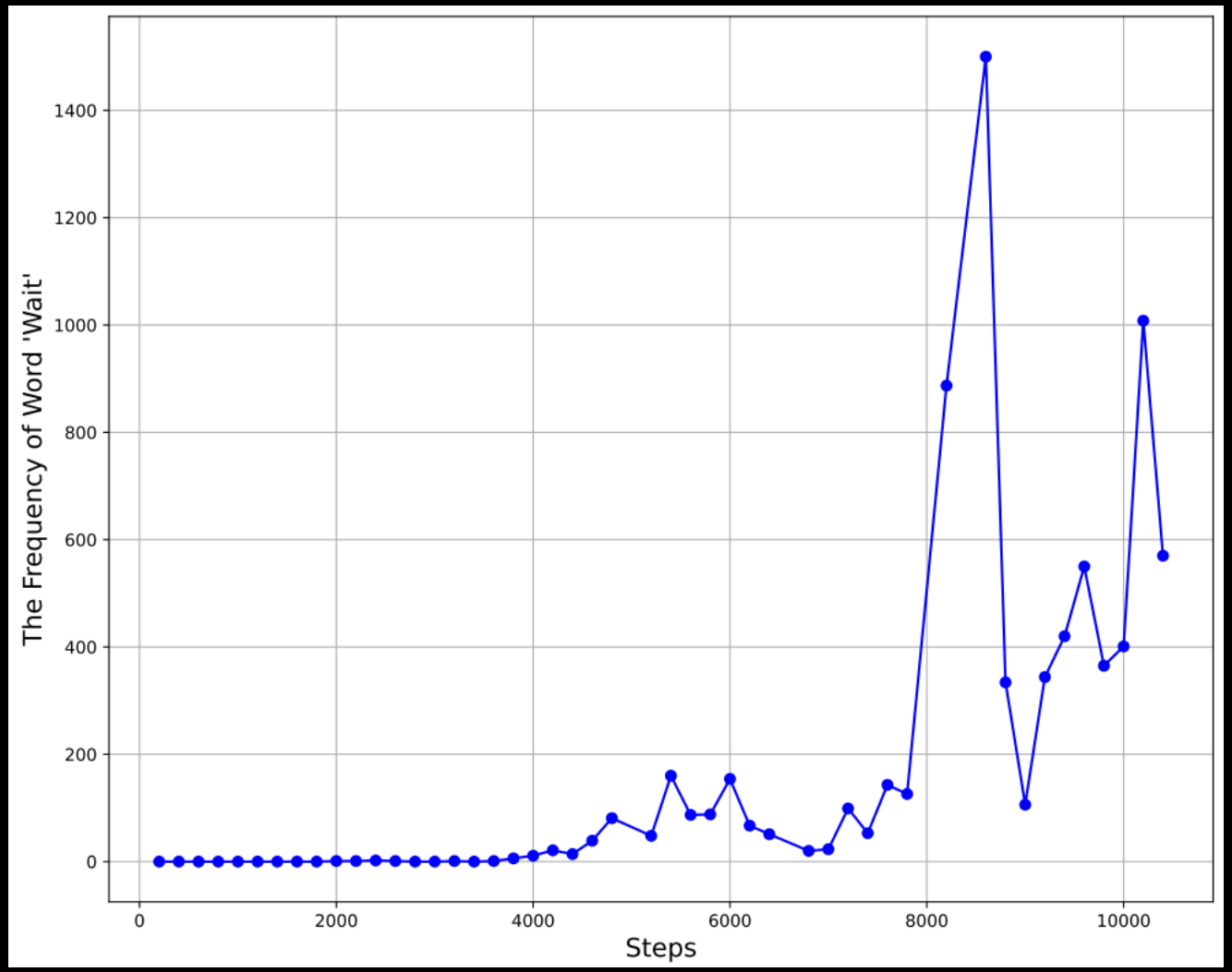
Next, I could square both sides again, treating the equation: $\cdots$

$\ldots$

The increased thinking time corresponds to advanced reasoning strategies such as self-reflection (e.g., instigated by "wait" word)!

DeepSeek-R1 paper. 2025

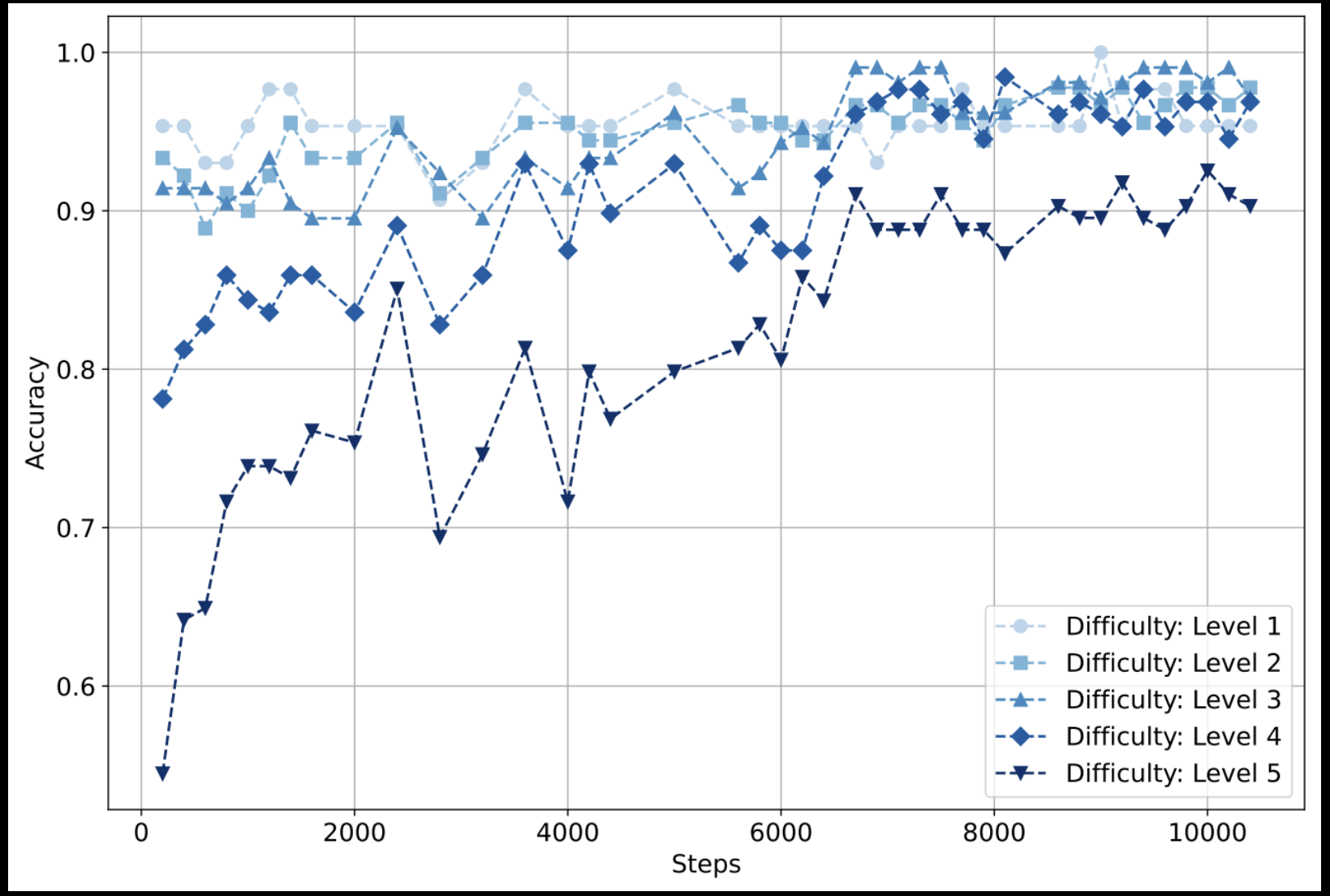
Analysis of RL Training



The increased thinking time corresponds to advanced reasoning strategies such as self-reflection (e.g., instigated by “wait” word)!

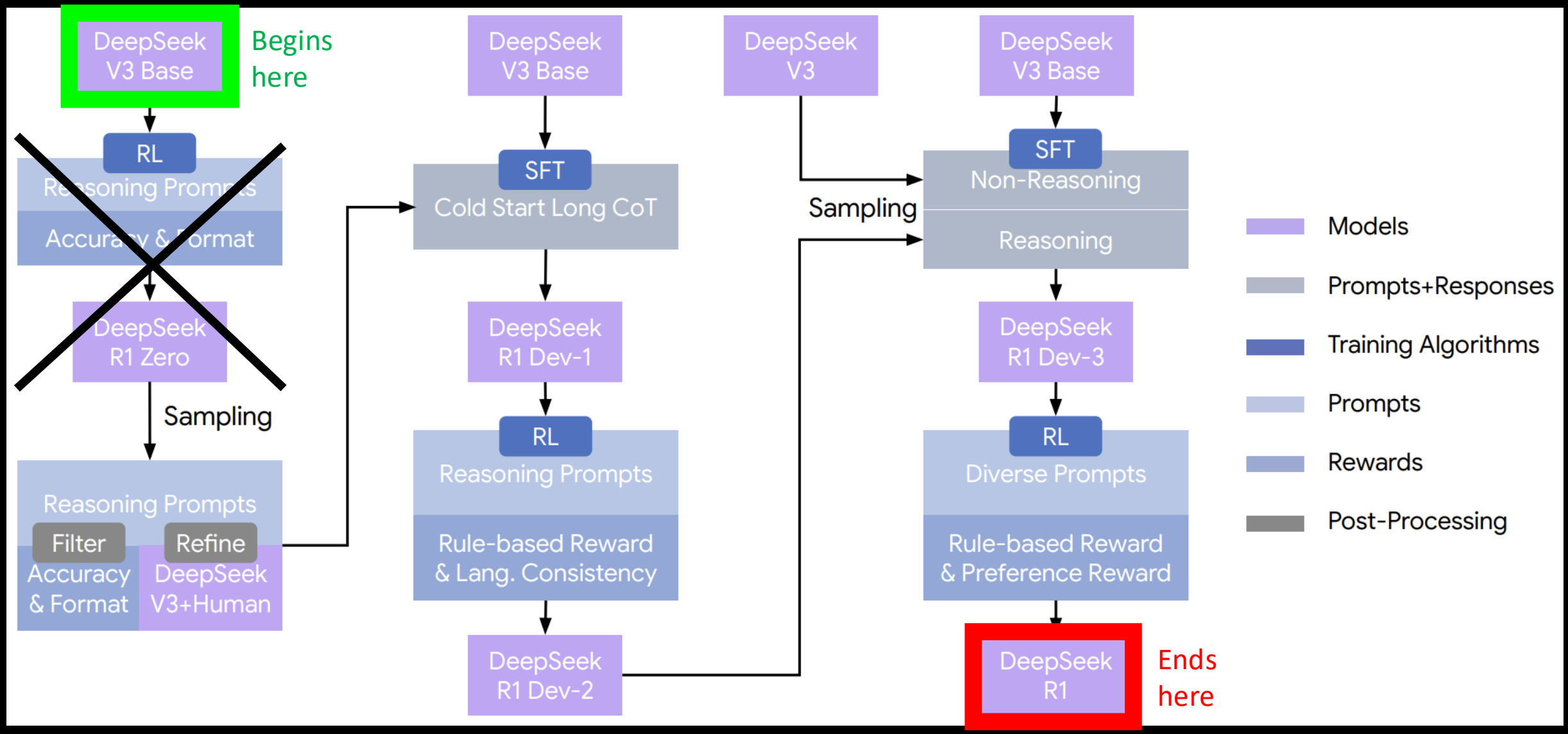
DeepSeek-R1 paper. 2025

Analysis of RL Training



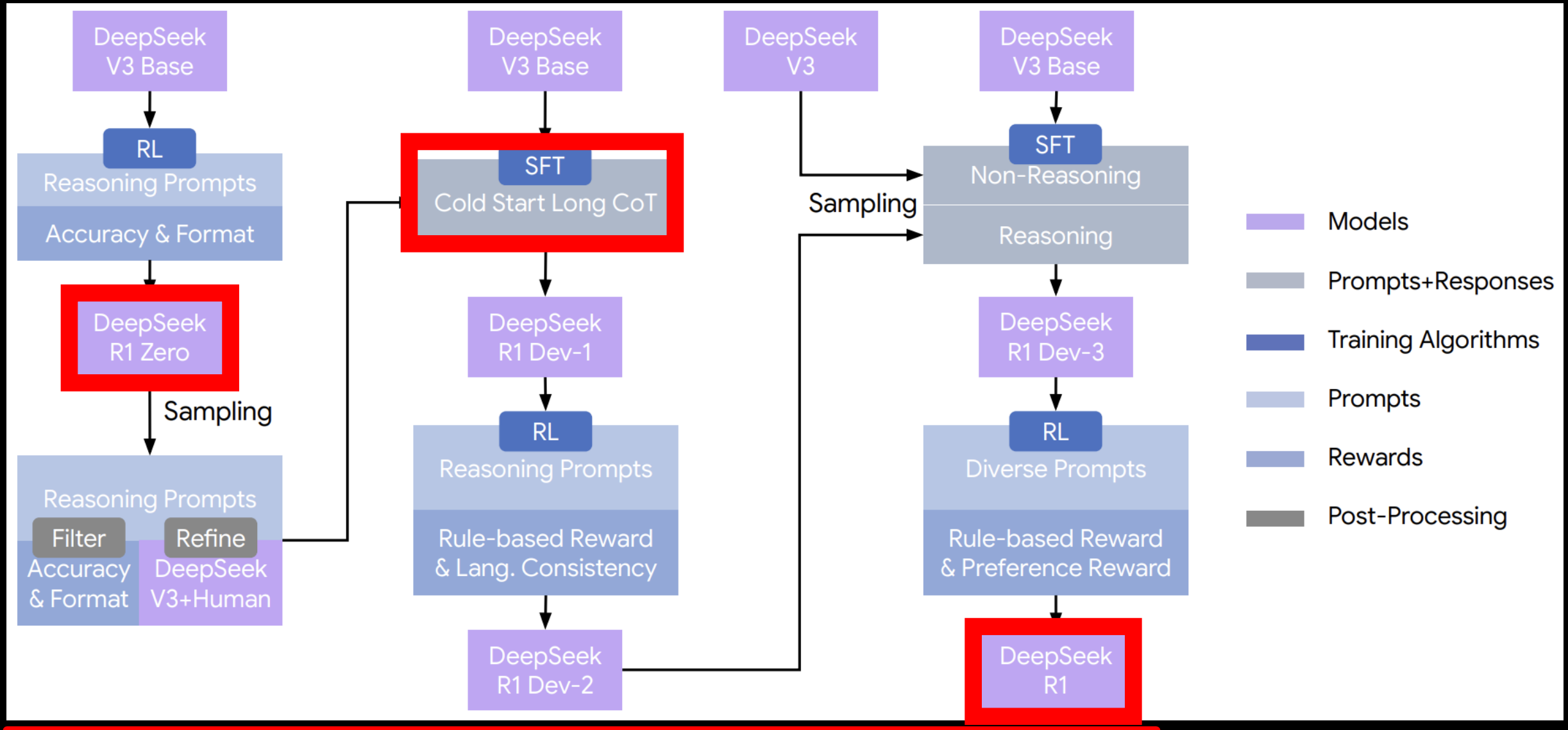
What are performance trends for different math problems with different difficulty levels as the training time grows?

DeepSeek-R1 paper. 2025



Remaining steps improve response readability, including to adopt typical conversational styles of humans in one language

DeepSeek-R1 paper. 2025



Training Costs	DeepSeek-R1-Zero	SFT data creation	DeepSeek-R1	Total
in H800 GPU Hours	101K	5K	41K	147K
in USD	$202K	$10K	$82K	$294K

DeepSeek-R1 paper. 2025



DeepSeek-V3

Mixture-of-experts + more architecture and HW tricks



DeepSeek-R1

Frontier-level reasoning with small training cost



DeepSeek-R1-Distill Series

Smaller variants

https://www.etsy.com/listing/1172948287/vintage-toys-collectible-miniature

Problem: Parameter-Heavy Models

Approximately how many parameters are in DeepSeek-R1?

(a) 670 million

(b) 6.7 billion

(c) 67 billion

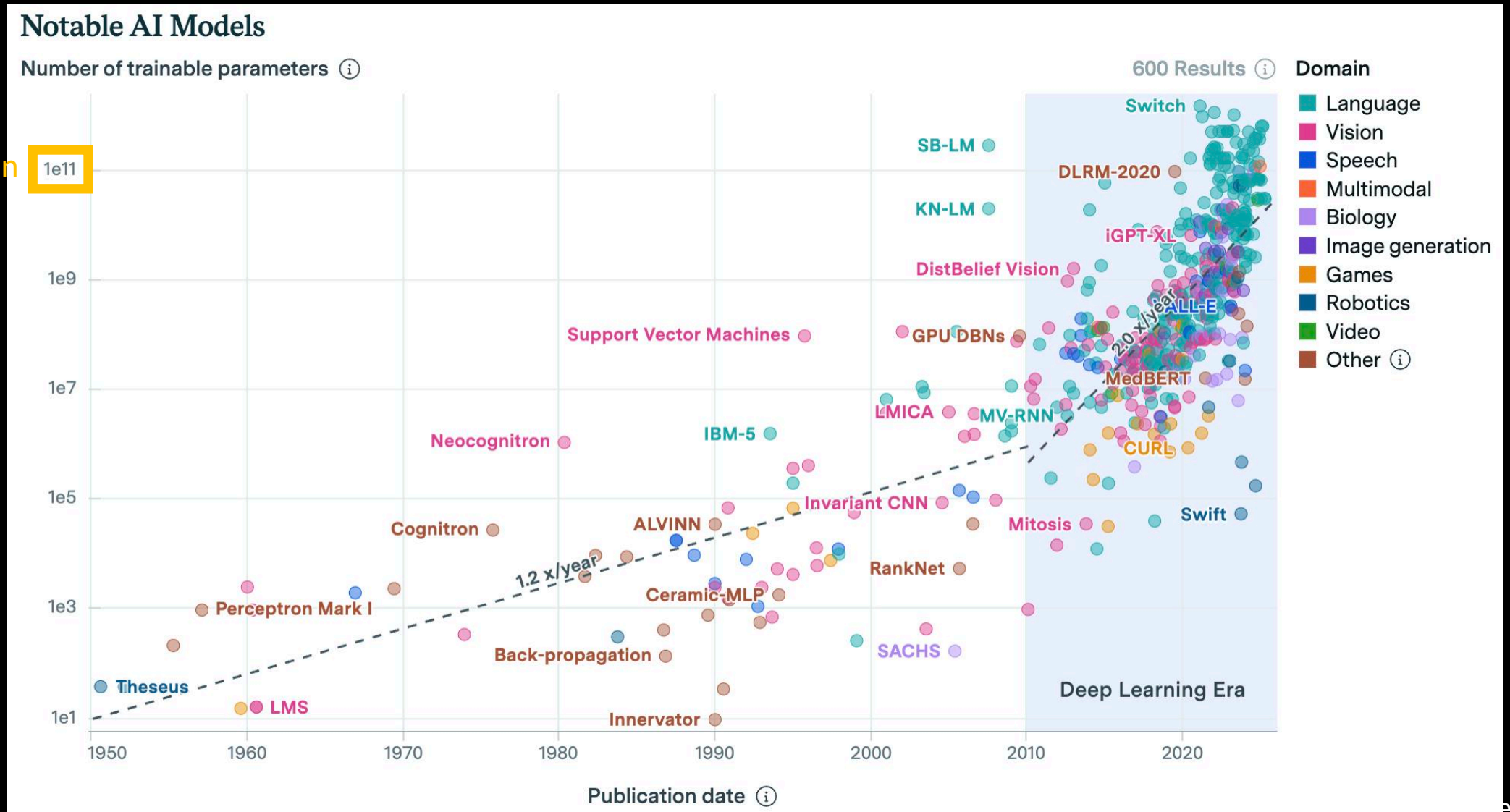
(d) 670 billion

(e) 6.7 trillion

https://en.wikipedia.org/wiki/GPT-4

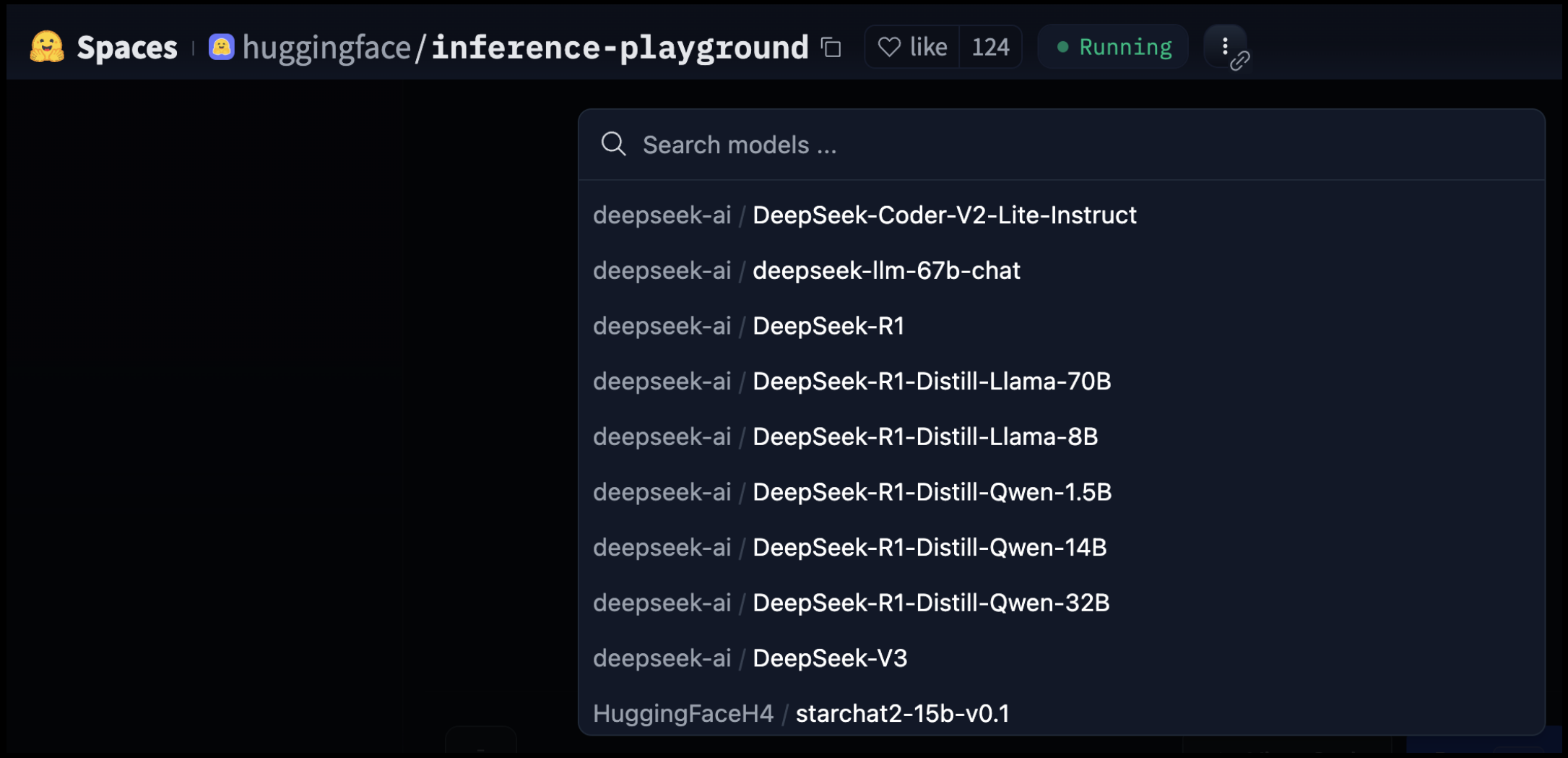
Problem: Parameter-Heavy Models

100 billion



explore-the-data

Today’s Status Quo: Multiple Model Sizes; e.g.,



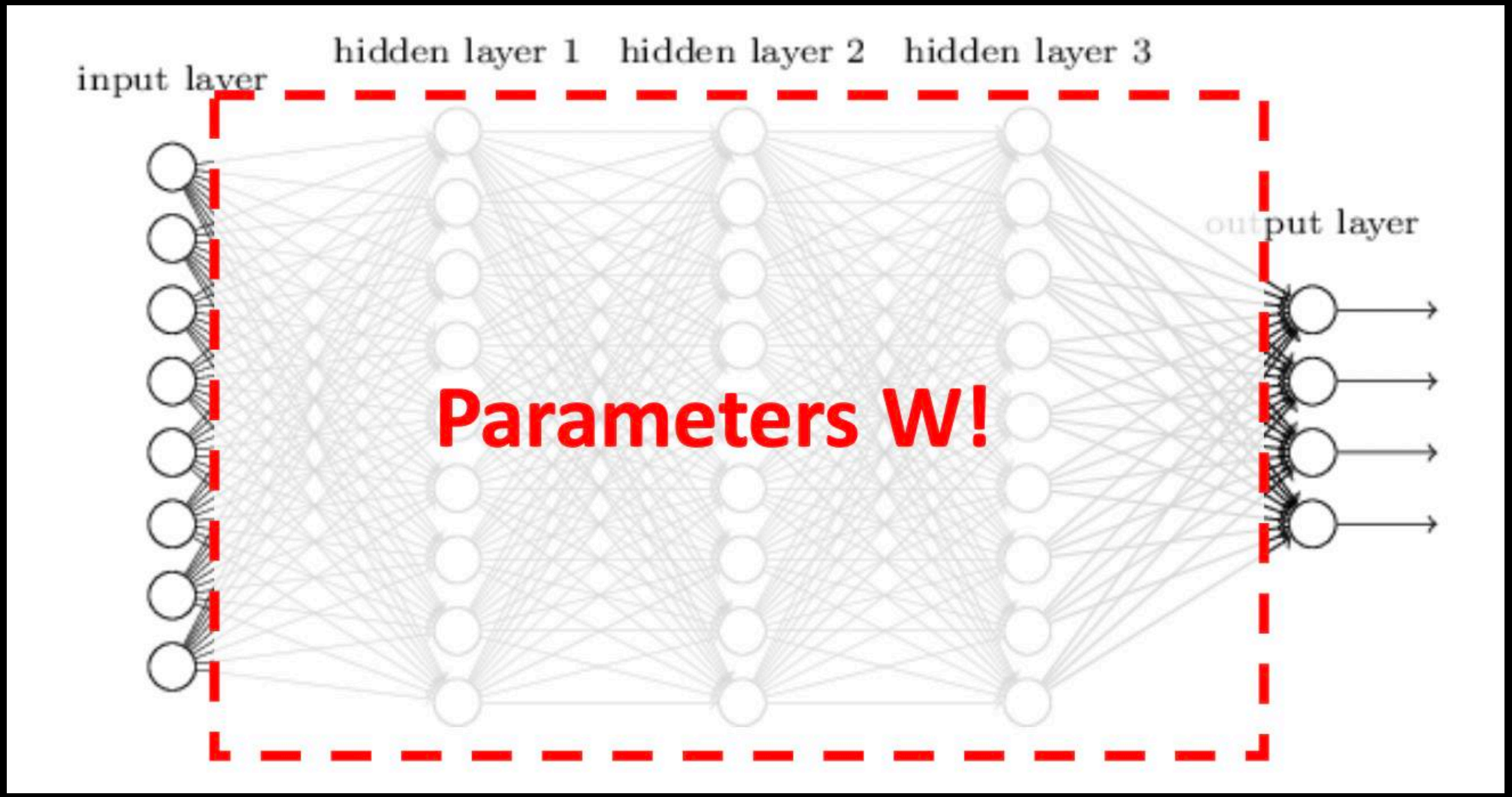
Background: Intuition



Have student learn from a knowledgeable teacher

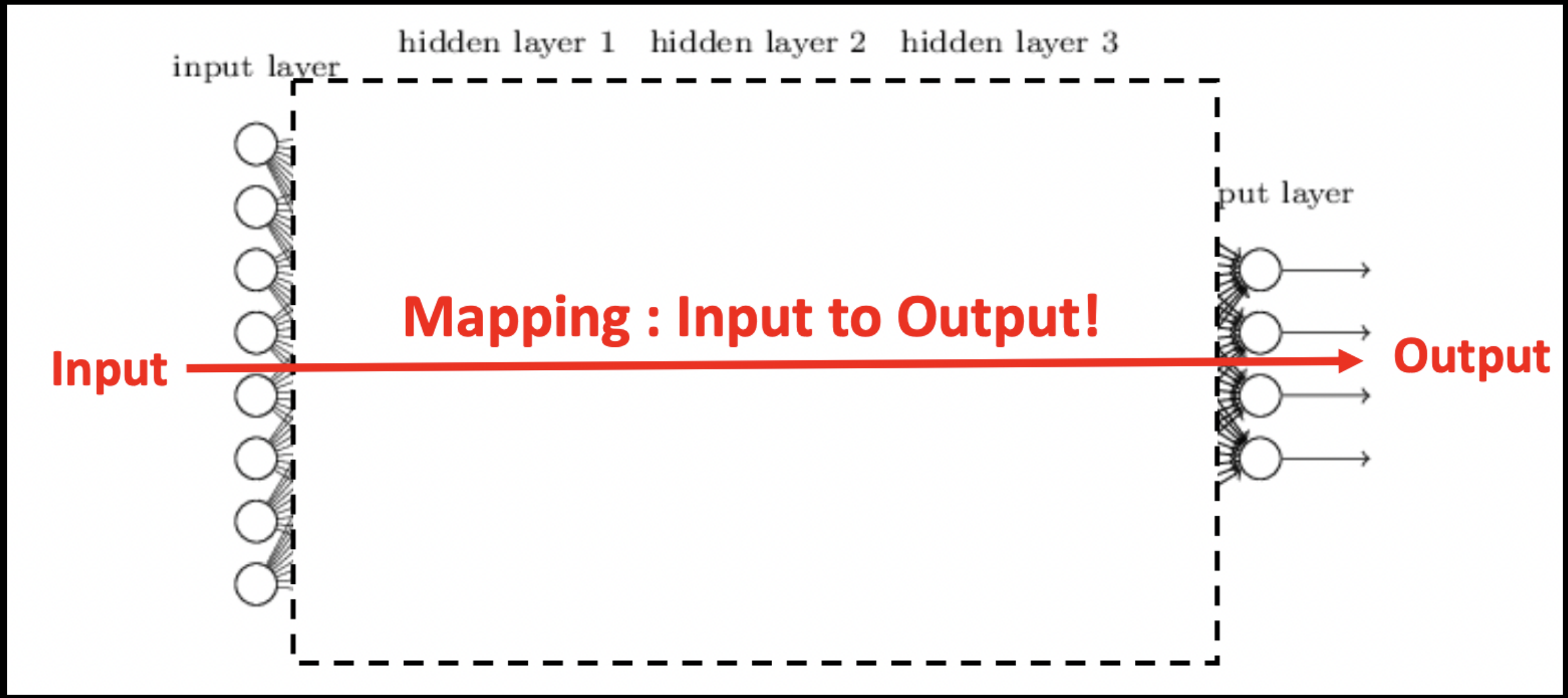
https://www.waterford.org/education/teacher-student-relationships/

Key Question: What is Knowledge?



http://ir.hit.edu.cn/~xiachongfeng/slides/Knowledge%20Distillation.pdf

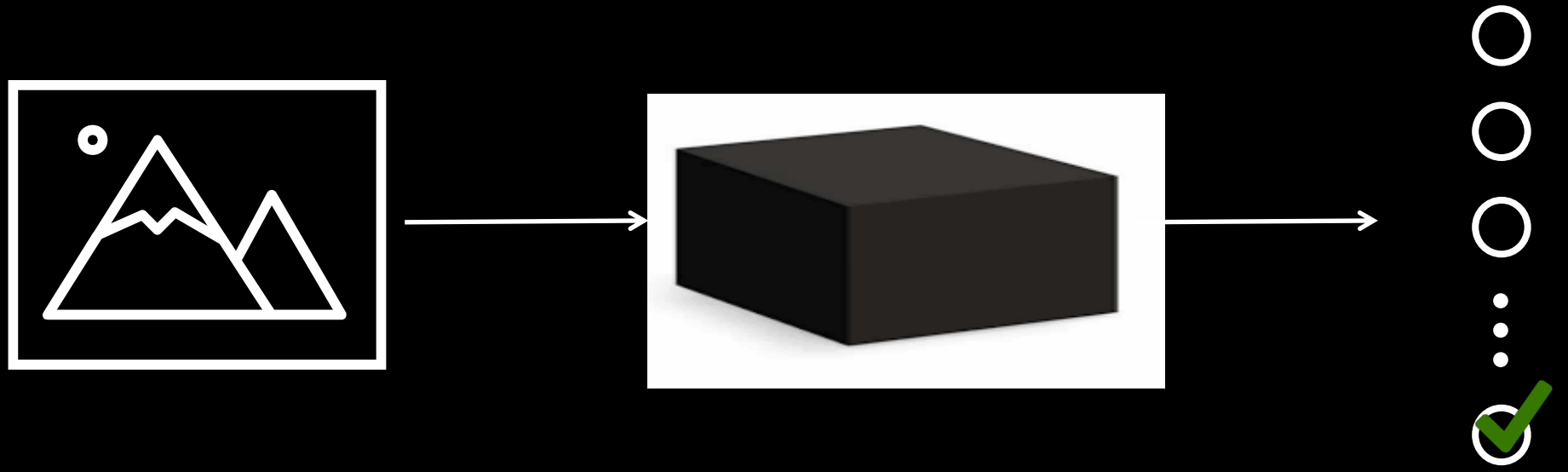
Knowledge Is: Input to Output Mapping



http://ir.hit.edu.cn/~xiachongfeng/slides/Knowledge%20Distillation.pdf

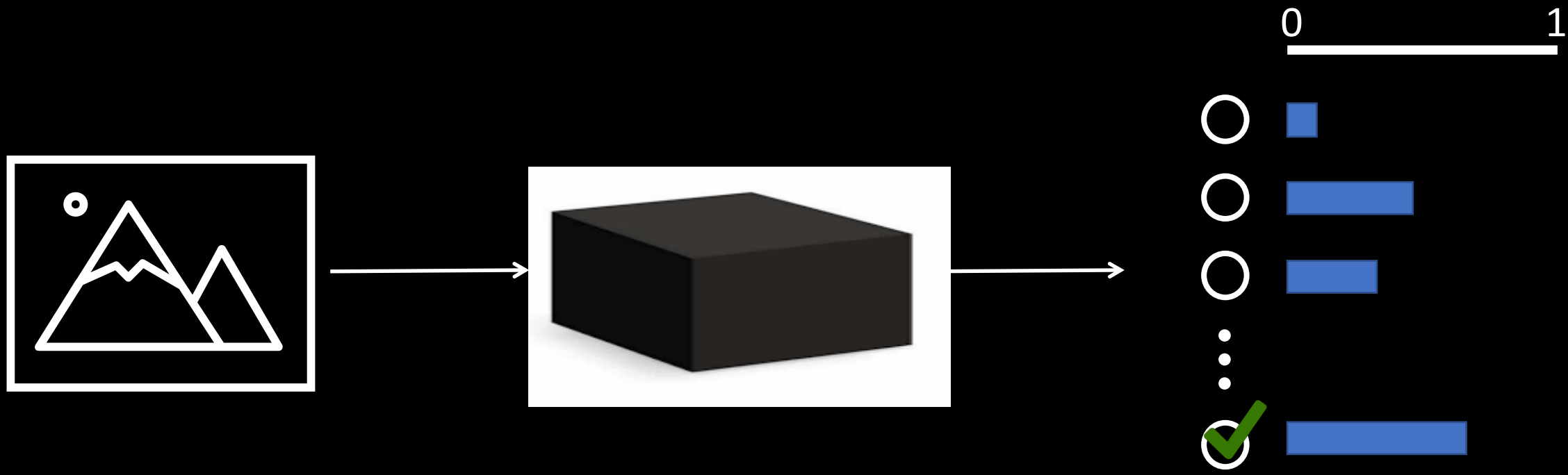
Knowledge Is: Input to Output Mapping

Target mapping: ground truth (1-hot vector)



Knowledge Is: Input to Output Mapping

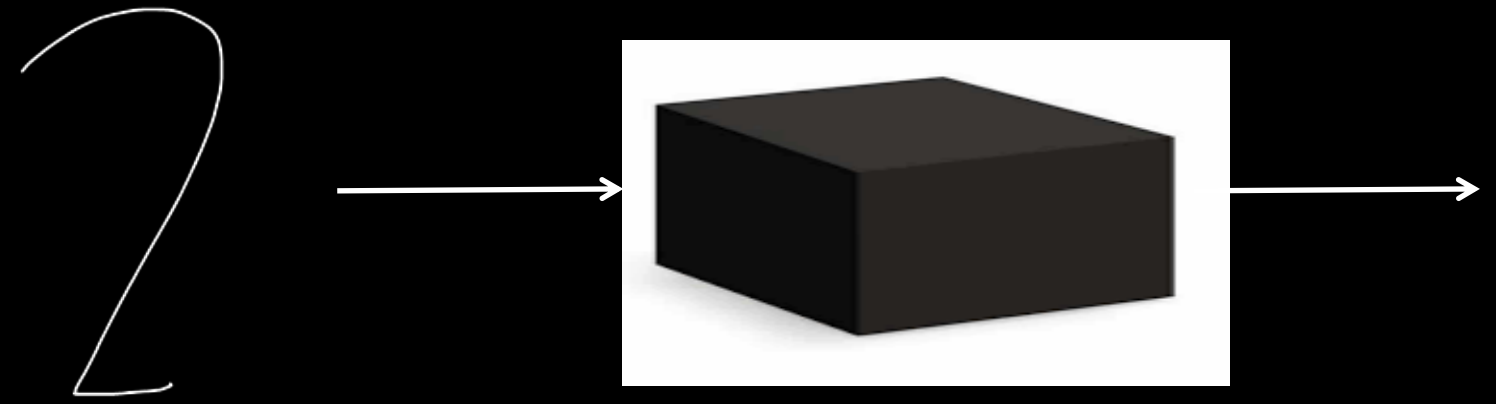
Target mapping: probability distribution from a model offers
further insights into similarities and differences of categories



Knowledge Is: Input to Output Mapping

Target mapping: probability distribution from a model offers further insights into similarities and differences of categories

- Attempts to identify ground truth category
- Also, shares that 2 has similar characteristics to 7 and 1

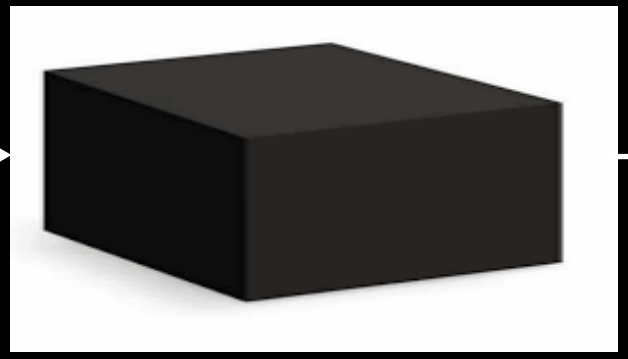


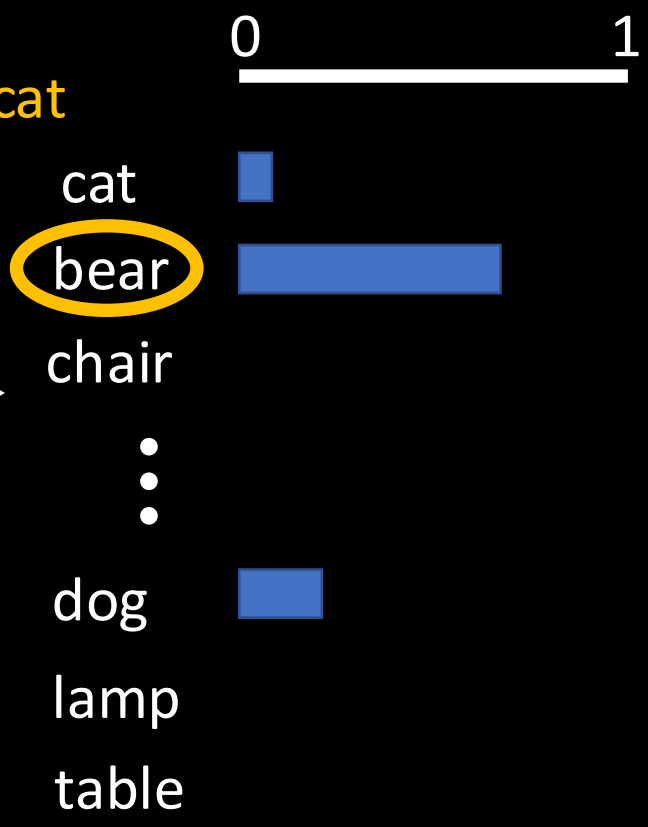
Knowledge Is: Input to Output Mapping

Target mapping: probability distribution from a model offers further insights into similarities and differences of categories

- Attempts to identify ground truth category
- Also, shares that bear has similar characteristics to dog and cat







Knowledge Is: Input to Output Mapping

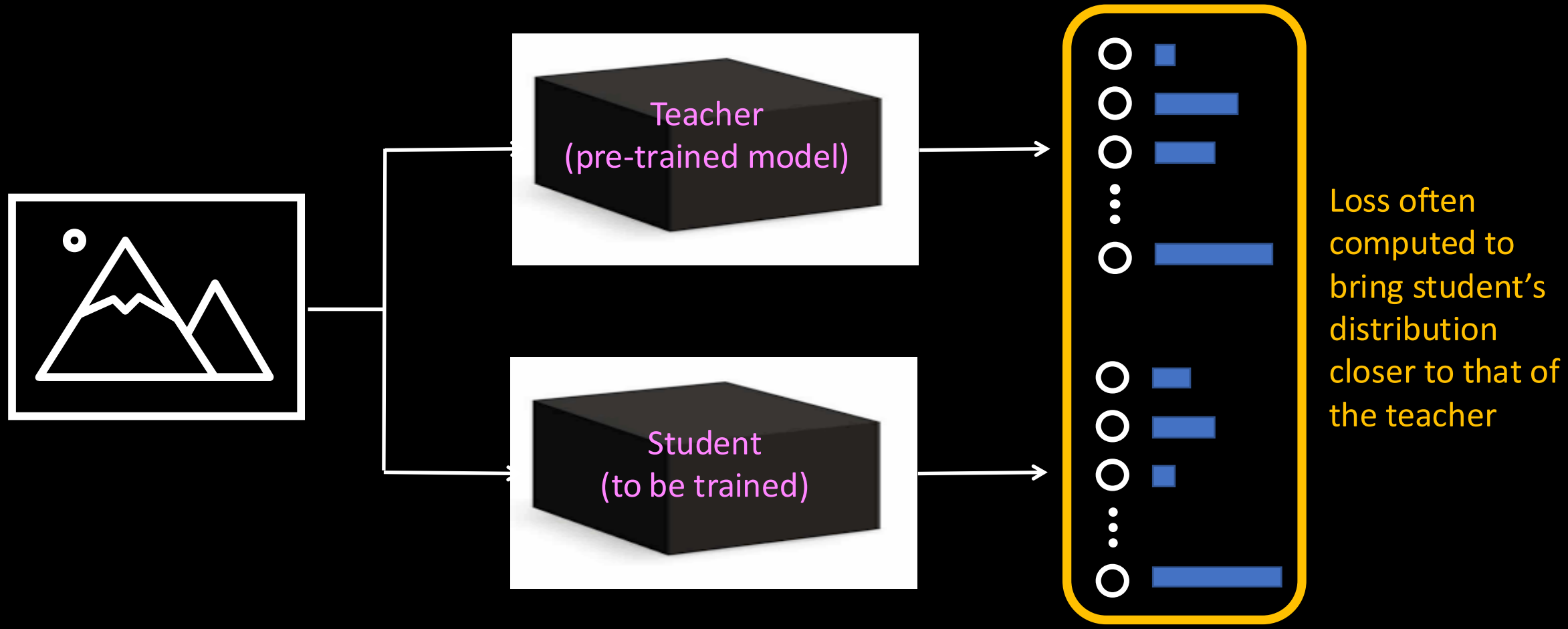
Target mapping: probability distribution from a model offers further insights into similarities and differences of categories

- Attempts to identify ground truth category
- Also, shares that bear has similar characteristics to dog and cat

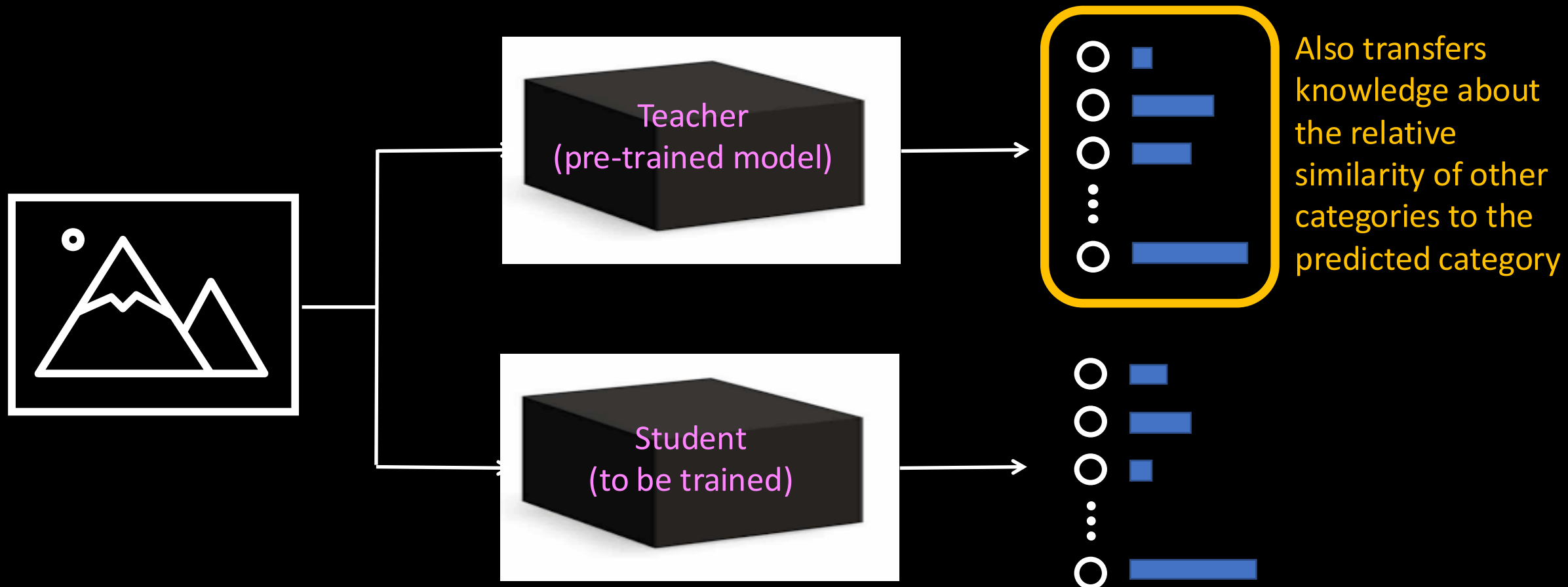
Classical idea: teach about ground truth and its relationships to other categories

Hinton, Vinyals, and Dean. Distilling the knowledge in a neural network. *arXiv* 2015.

Classic Knowledge Distillation: Teach Student the “Dark Knowledge” of Teacher

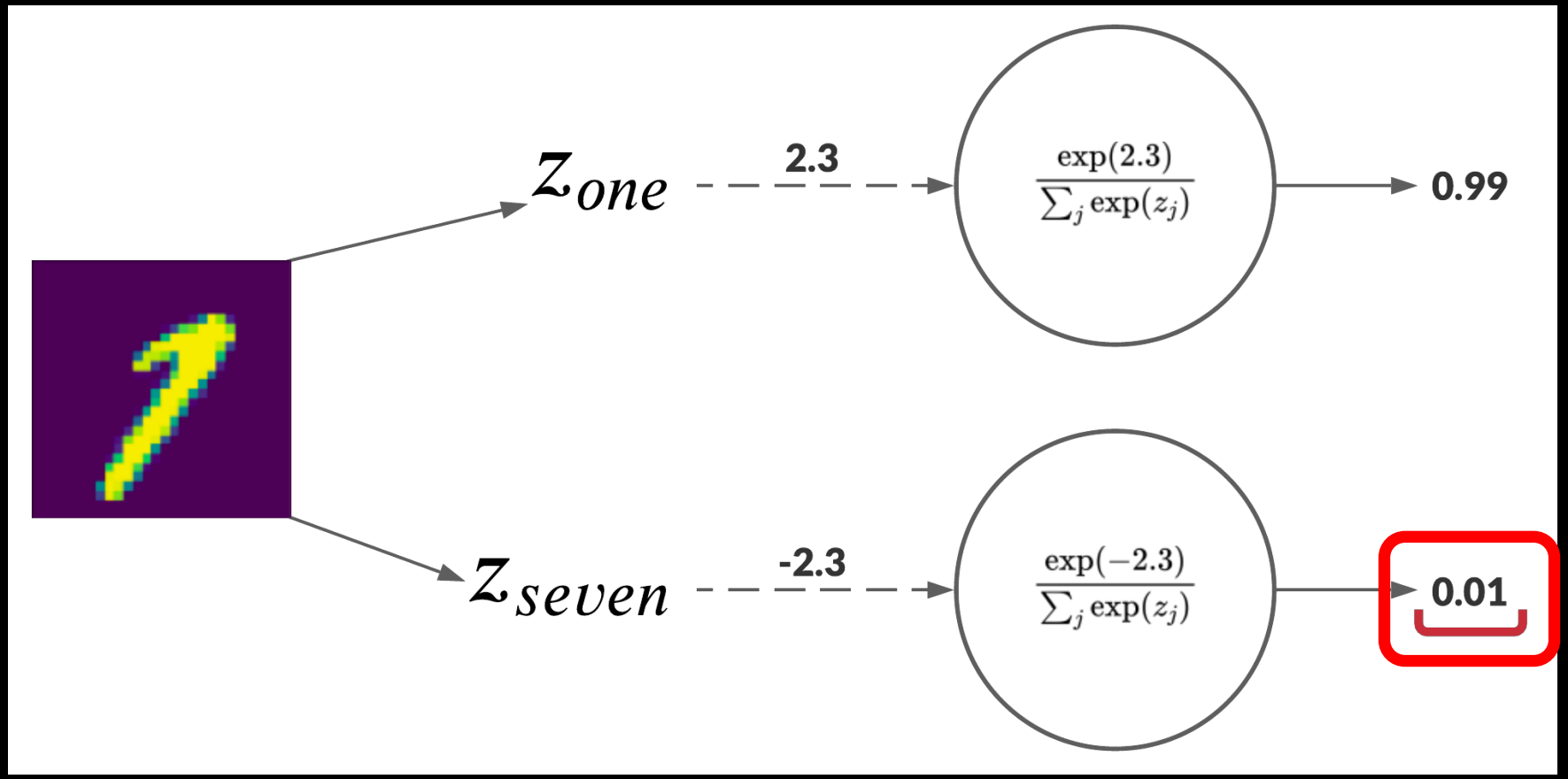


Classic Knowledge Distillation: Teach Student the “Dark Knowledge” of Teacher



Classic Knowledge Distillation: Rebalance ("Soften") Probability Distribution Across Categories

Recall Softmax: converts scores into a probability distribution that sums to 1



Want to enhance knowledge of this relationship

https://wandb.ai/authors/knowledge-distillation/reports/Distilling-Knowledge-in-Neural-Networks--VmlldzoyMjkxODk

Classic Knowledge Distillation: Rebalance ("Soften") Probability Distribution Across Categories

Generalized Softmax: converts scores into a probability distribution summing to 1, with temperature

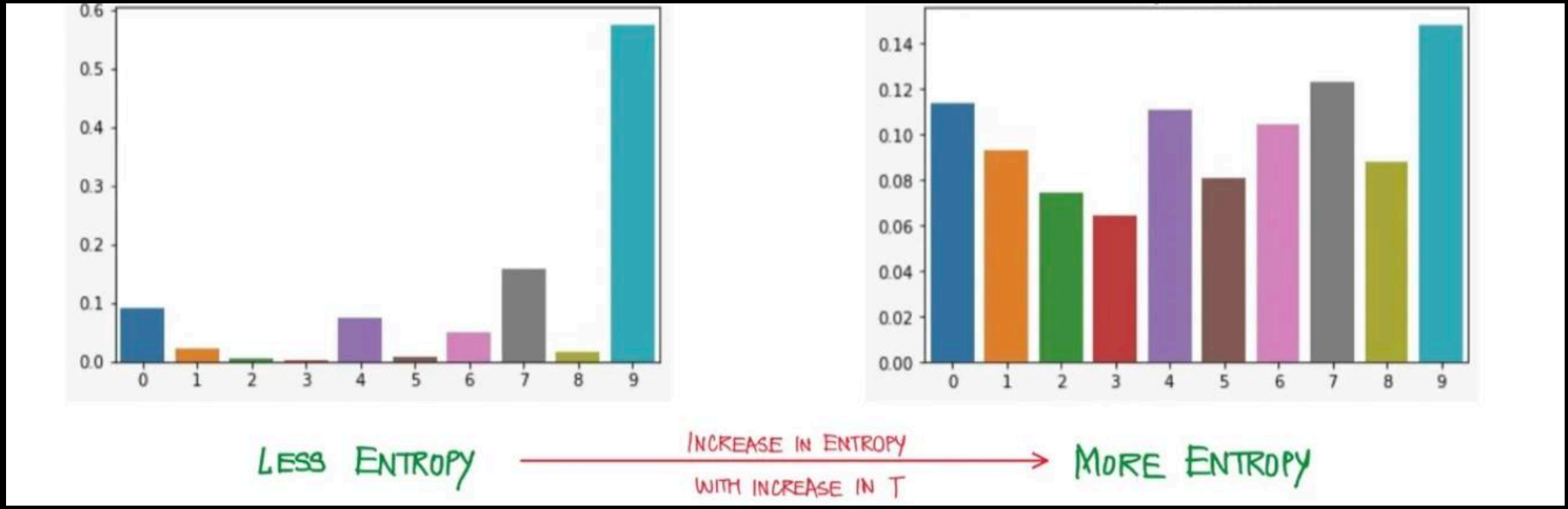
$$\sigma(\mathbf{z})_i = \frac{\exp(z_i / T)}{\sum_j \exp(z_j / T)}$$

What is the typical value of T used for softmax?

Idea: set the temperature to a value greater than 1

Classic Knowledge Distillation: Rebalance ("Soften") Probability Distribution Across Categories

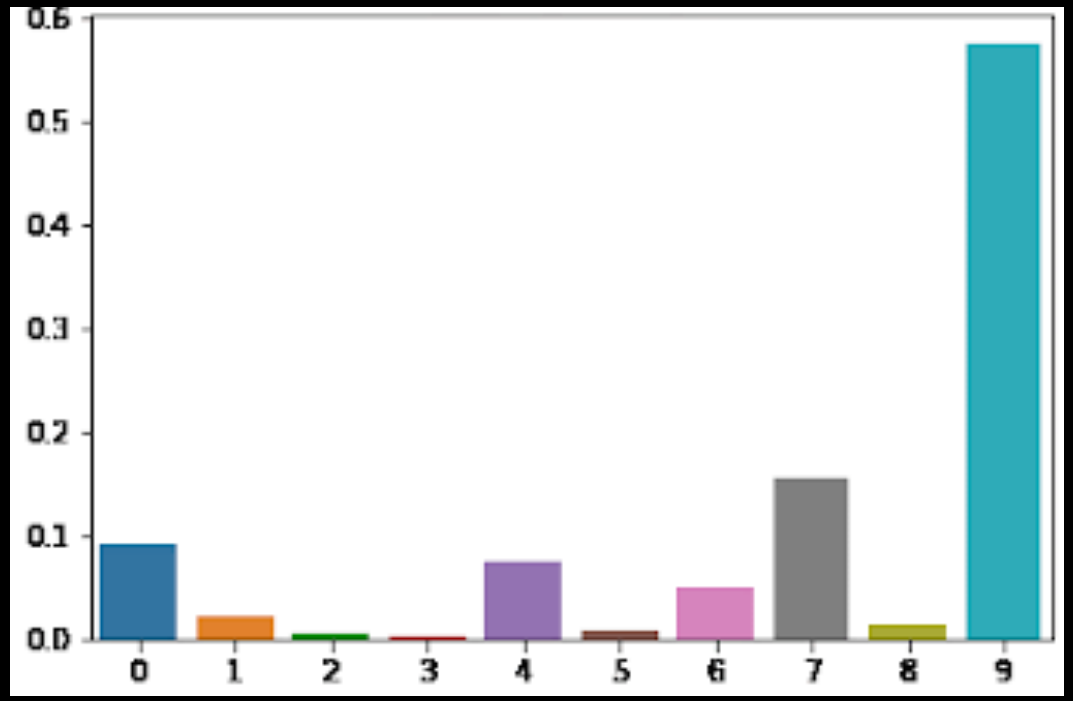
Generalized Softmax: converts scores into a probability distribution summing to 1, with temperature



https://medium.com/@harshit158/softmax-temperature-5492e4007f71

Classic Knowledge Distillation: Rebalance (“Soften”) Probability Distribution Across Categories

Generalized Softmax: converts scores into a probability distribution summing to 1, with temperature

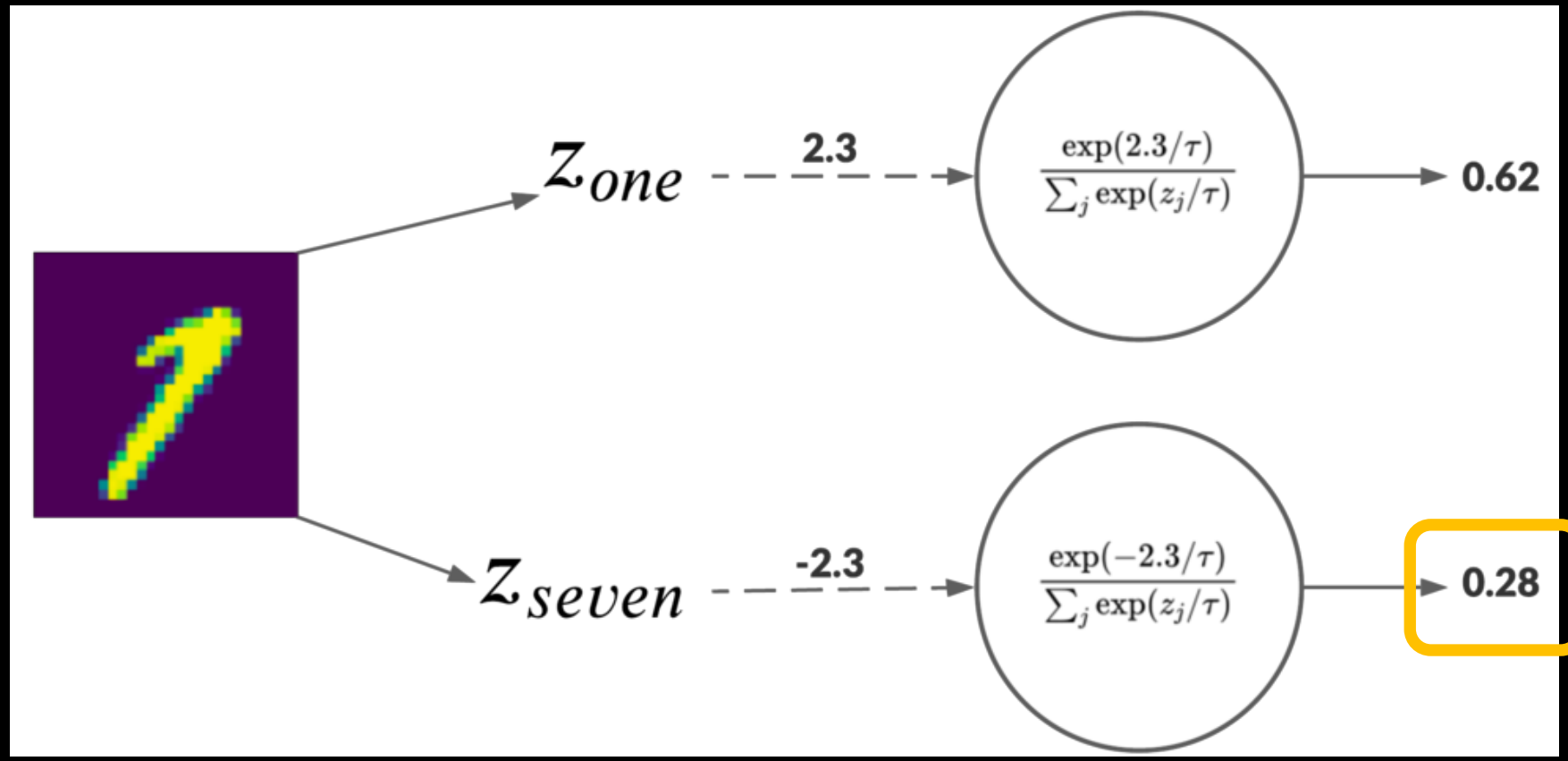


Larger T values provides greater information about which categories are similar to the teacher’s predicted category

https://medium.com/@harshit158/softmax-temperature-5492e4007f71

Classic Knowledge Distillation: Rebalance ("Soften") Probability Distribution Across Categories

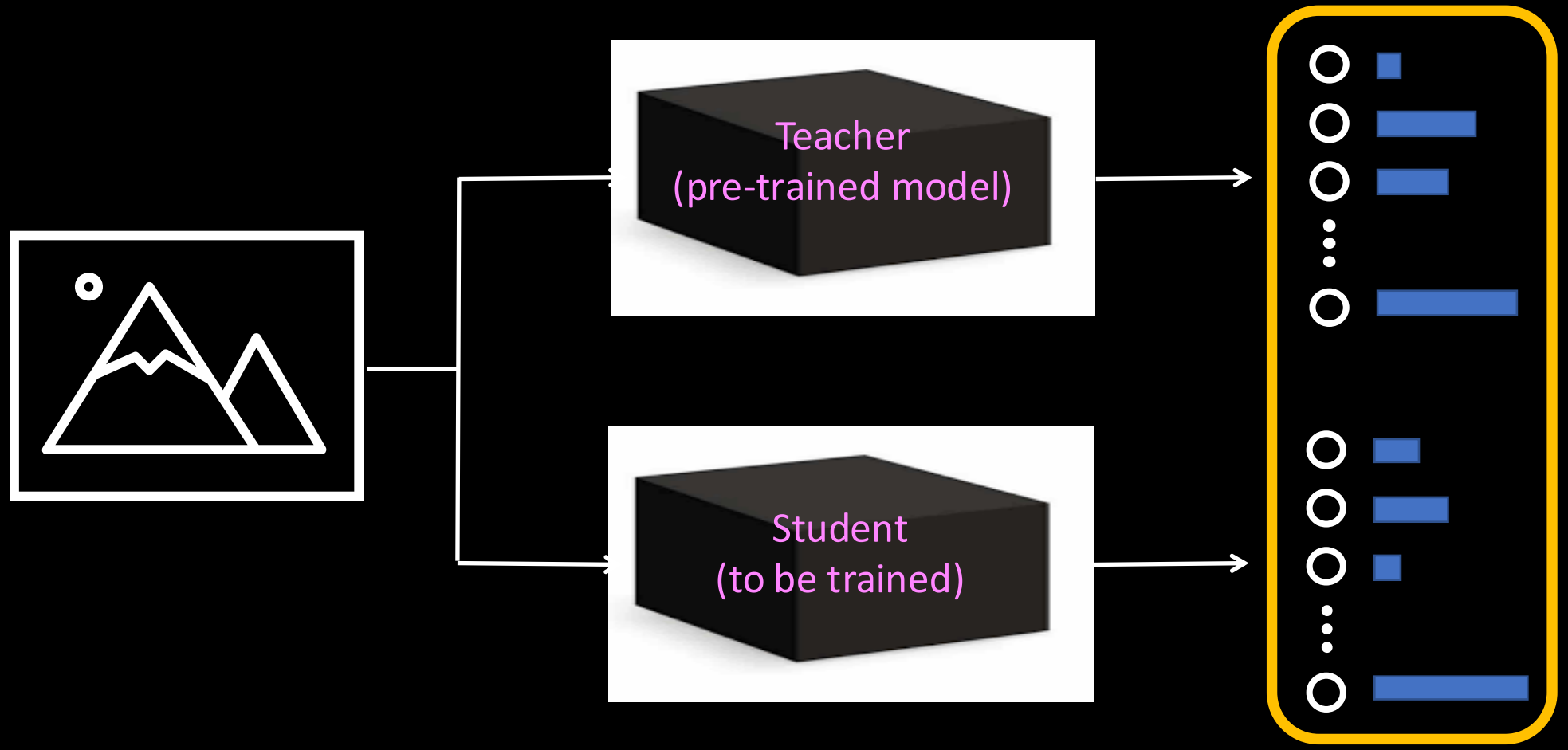
Setting temperature=5



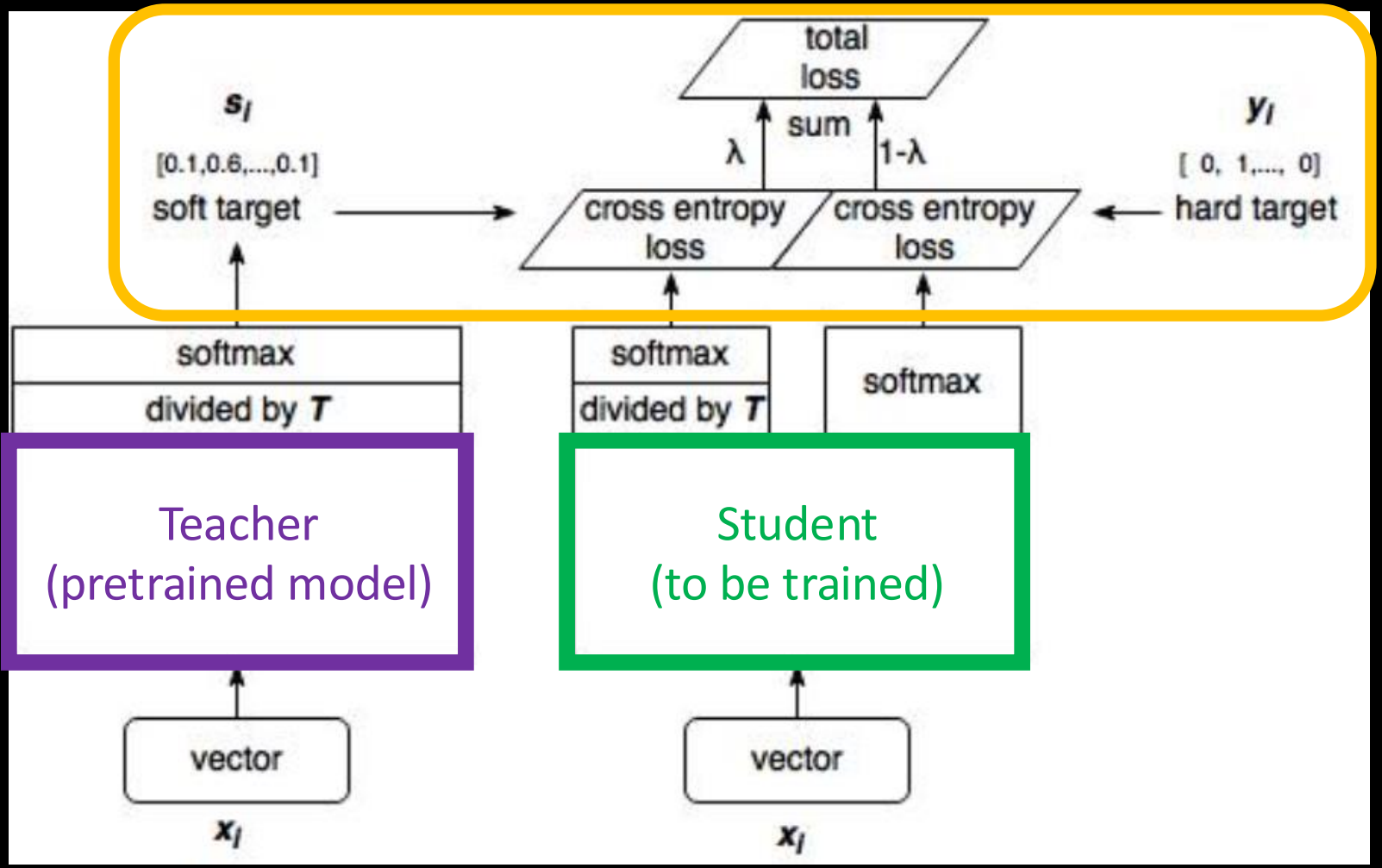
Knowledge of this relationship enhanced from 0.01

https://wandb.ai/authors/knowledge-distillation/reports/Distilling-Knowledge-in-Neural-Networks--VmlldzoyMjkxODk

Classic Knowledge Distillation: Teach Student the “Dark Knowledge” of Teacher



Classic Knowledge Distillation: Teach Student the "Dark Knowledge" of Teacher

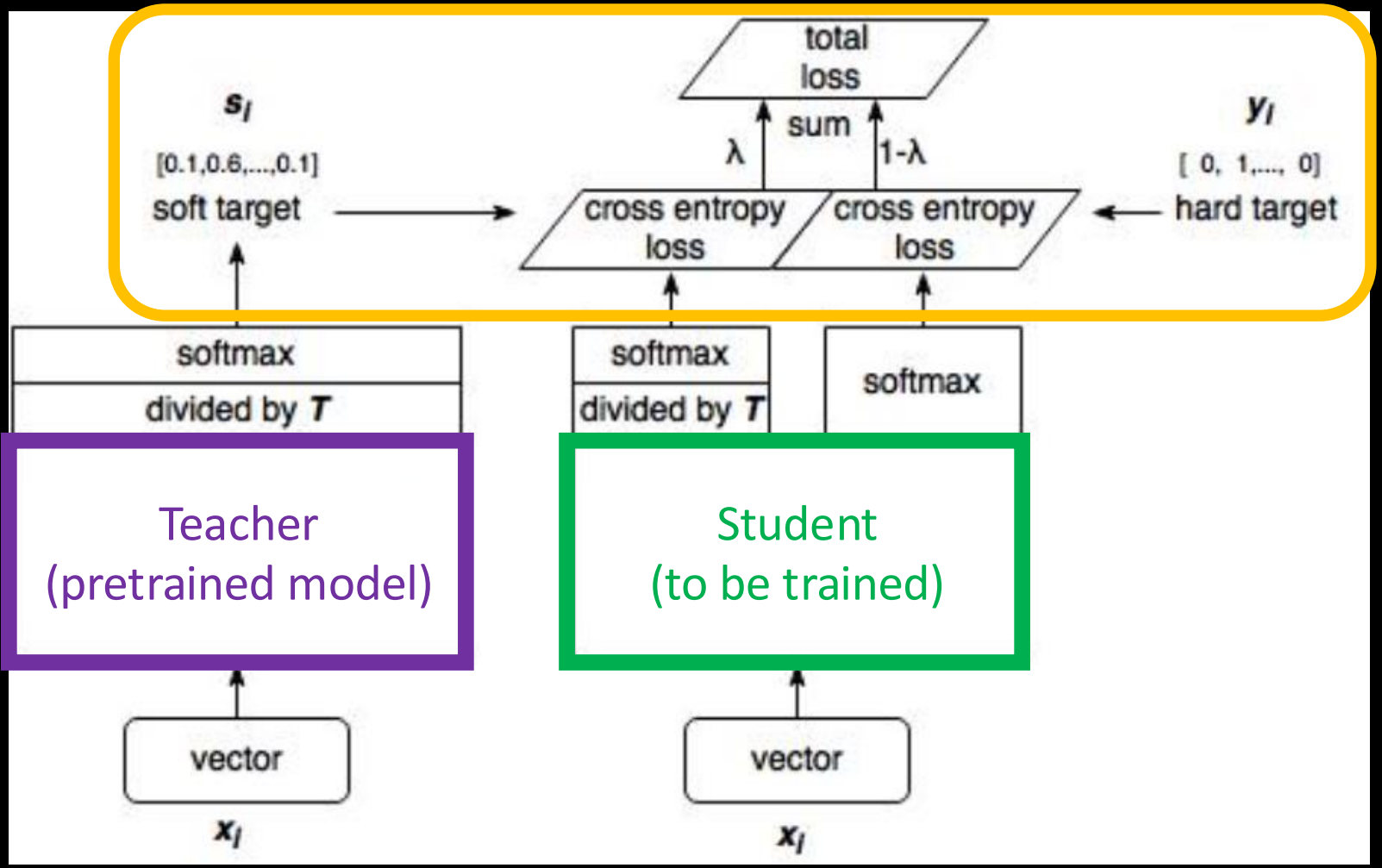


Total loss computed during training is a weighted sum of the conventional cross entropy loss and the "distillation loss"

At test time, only the student model with softmax is used

https://blog.csdn.net/qq_22749699/article/details/79460817

DeepSeek's KD Variants: Supervise Only With the Teacher's Output to Mimic its Reasoning Process (No Extra Loss Terms to Match Prob. Distribution)



https://blog.csdn.net/qq_22749699/article/details/79460817

DeepSeek's KD Variants: Supervise Only With the Teacher's Output to Mimic its Reasoning Process (No Extra Loss Terms to Match Prob. Distribution)

Distilled Model	Base Model
DeepSeek-R1-Distill-Qwen-1.5B	Qwen2.5-Math-1.5B
DeepSeek-R1-Distill-Qwen-7B	Qwen2.5-Math-7B
DeepSeek-R1-Distill-Qwen-14B	Qwen2.5-14B
DeepSeek-R1-Distill-Qwen-32B	Qwen2.5-32B
DeepSeek-R1-Distill-Llama-8B	Llama-3.1-8B
DeepSeek-R1-Distill-Llama-70B	Llama-3.3-70B-Instruct

Better performance achieved using SFT with distillation than training models from scratch

DeepSeek-R1 paper. 2025

KD Ethics?

What's your take on the ethics of this?

The AI Cold War? US tech companies accuse China's AI firms of stealing billions in research

So-called distillation attacks gather responses from AI models to teach smaller ones.

As the United States and China race to develop artificial intelligence (AI), American firm Anthropic is the latest company to sound the alarm that Chinese AI companies have been stealing the technology that could decide who wins.

DeepSeek, Moonshot AI and MiniMax secretly generated over 16 million conversations with Anthropic's AI chatbot Claude, using more than 24,000 fake accounts, to harvest its intelligence and train their own competing models, the company alleges.

OpenAI and Google have also warned about similar accusations at Chinese firms this month, raising fears that China is short-circuiting years of costly AI research.

What is AI distillation?

Model extraction attacks (MEA), otherwise known as "distillation", is a technique in which someone with access to a powerful AI model uses it to train a cheaper, faster rival.

The method feeds the larger model thousands of questions, collects its answers, and uses those responses to teach a new model to think in the same way.

https://www.euronews.com/next/2026/02/26/the-ai-cold-war-us-tech-companies-accuse-chinas-ai-firms-of-stealing-billions-in-research



DeepSeek-V3

Mixture-of-experts + more architecture and HW tricks



DeepSeek-R1

Frontier-level reasoning with small training cost



DeepSeek-R1-Distill Series

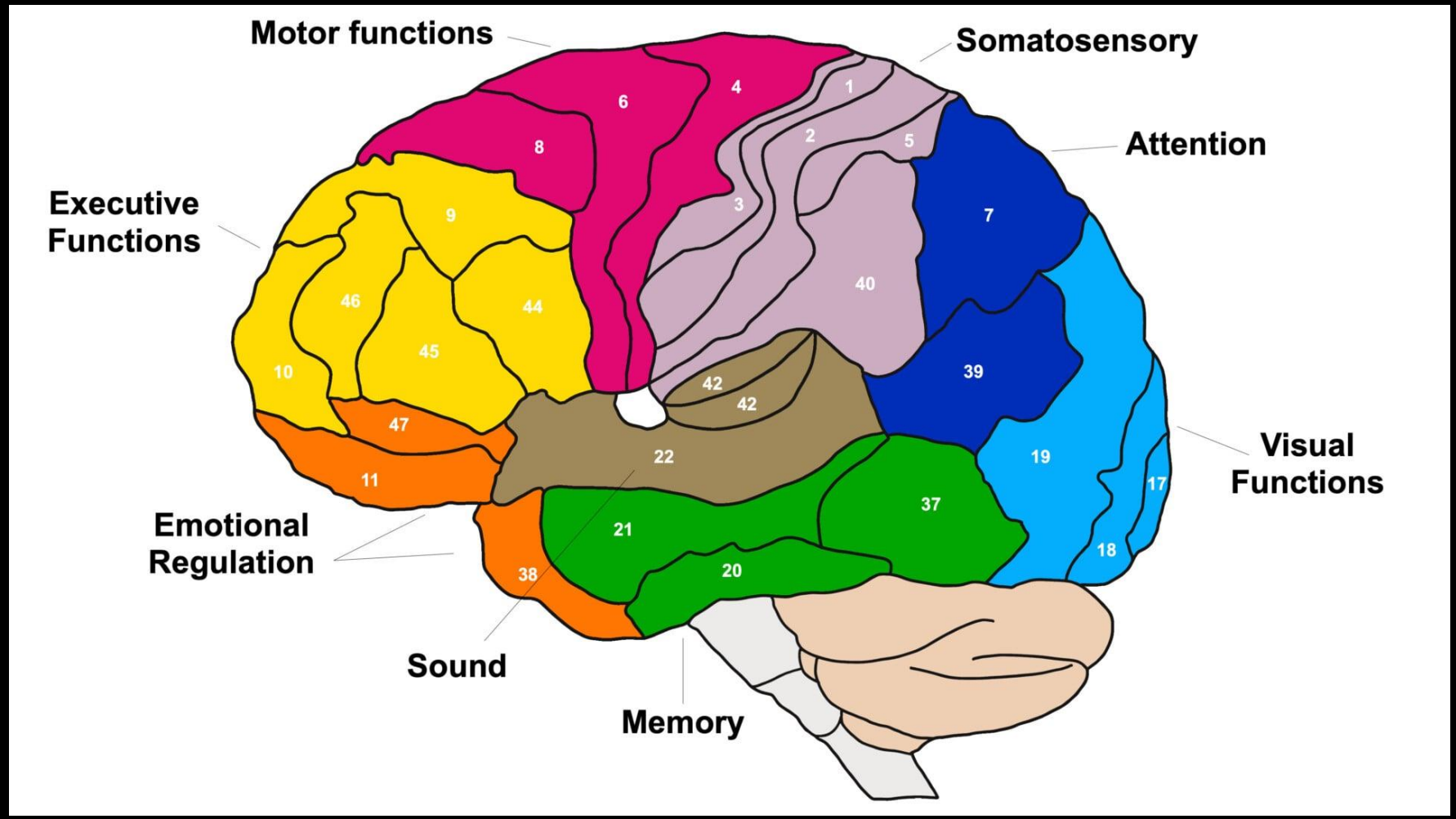
Smaller variants

https://www.etsy.com/listing/1172948287/vintage-toys-collectible-miniature

Ideas

- Mixture of experts: reduce computation
- Multi-head latent attention: reduce computation
- Multi-token prediction: accelerate computation

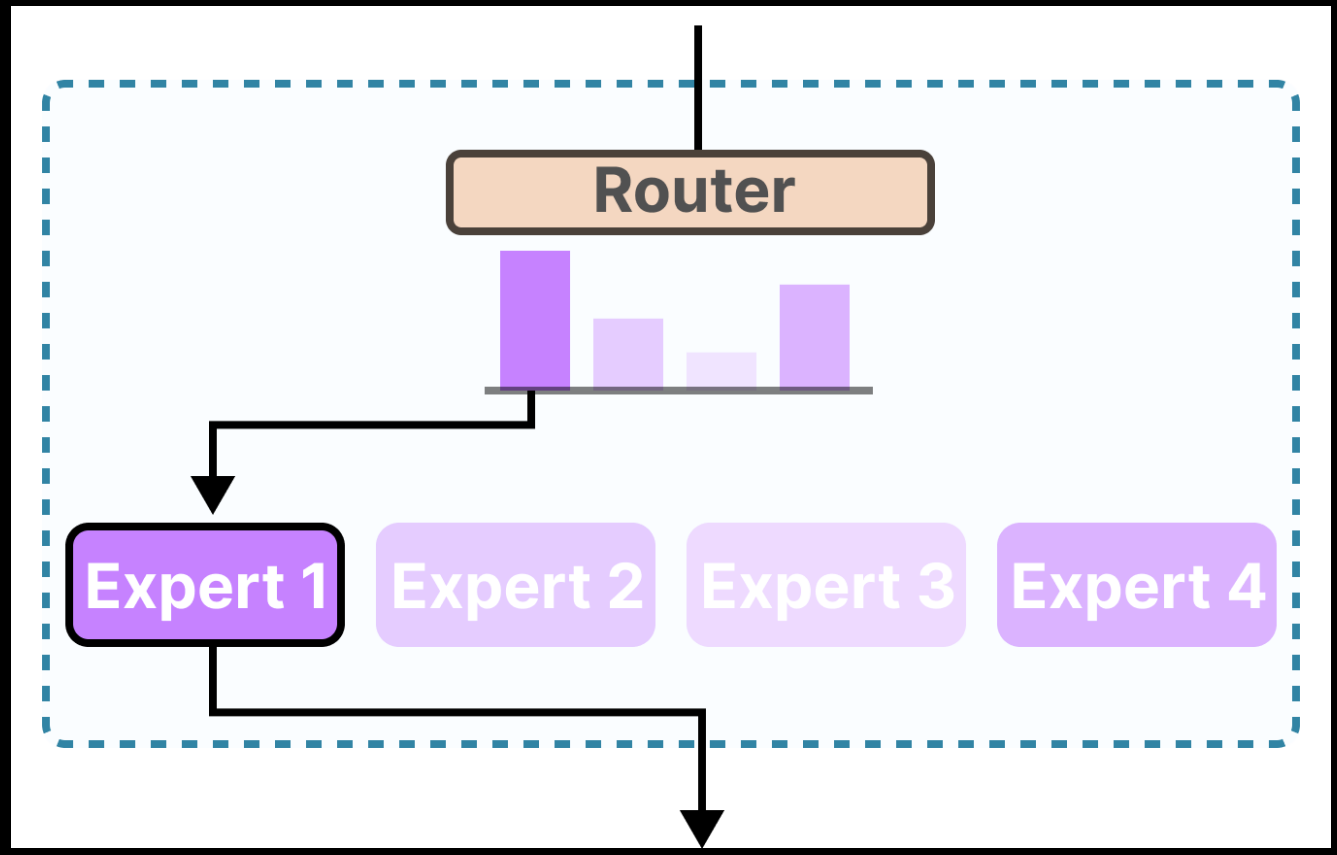
Idea: Only Use Part of the Network



https://www.simplypsychology.org/brodmann-areas.html

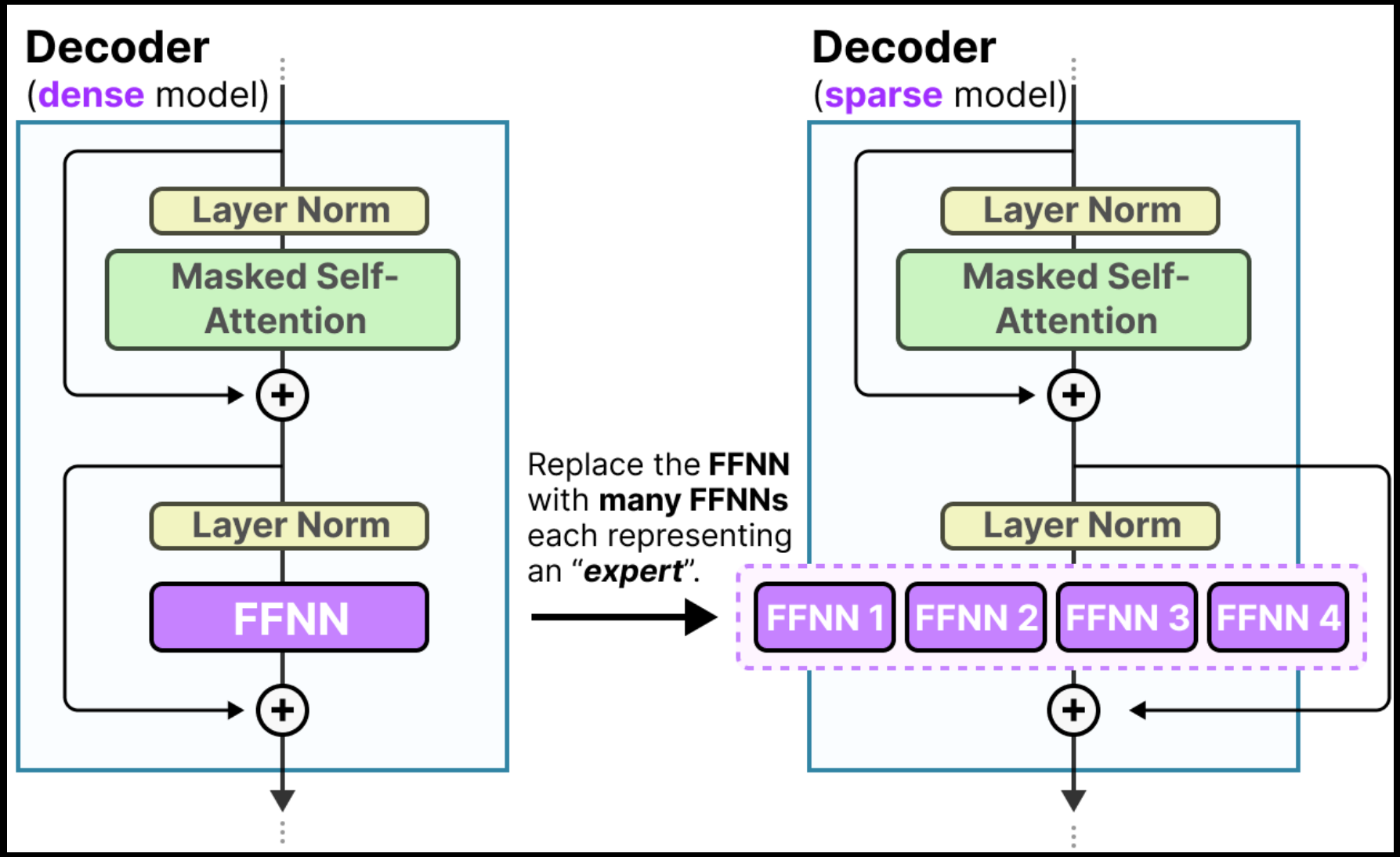
Mixture of Experts

- Activates a subset of model parameters at inference time



https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mixture-of-experts

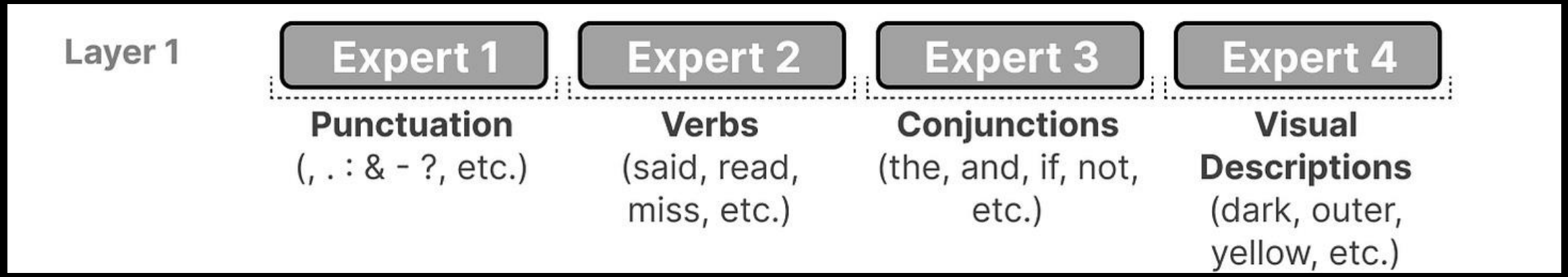
Mixture of Experts: Expert Implementation



https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mixture-of-experts

Mixture of Experts: Experts

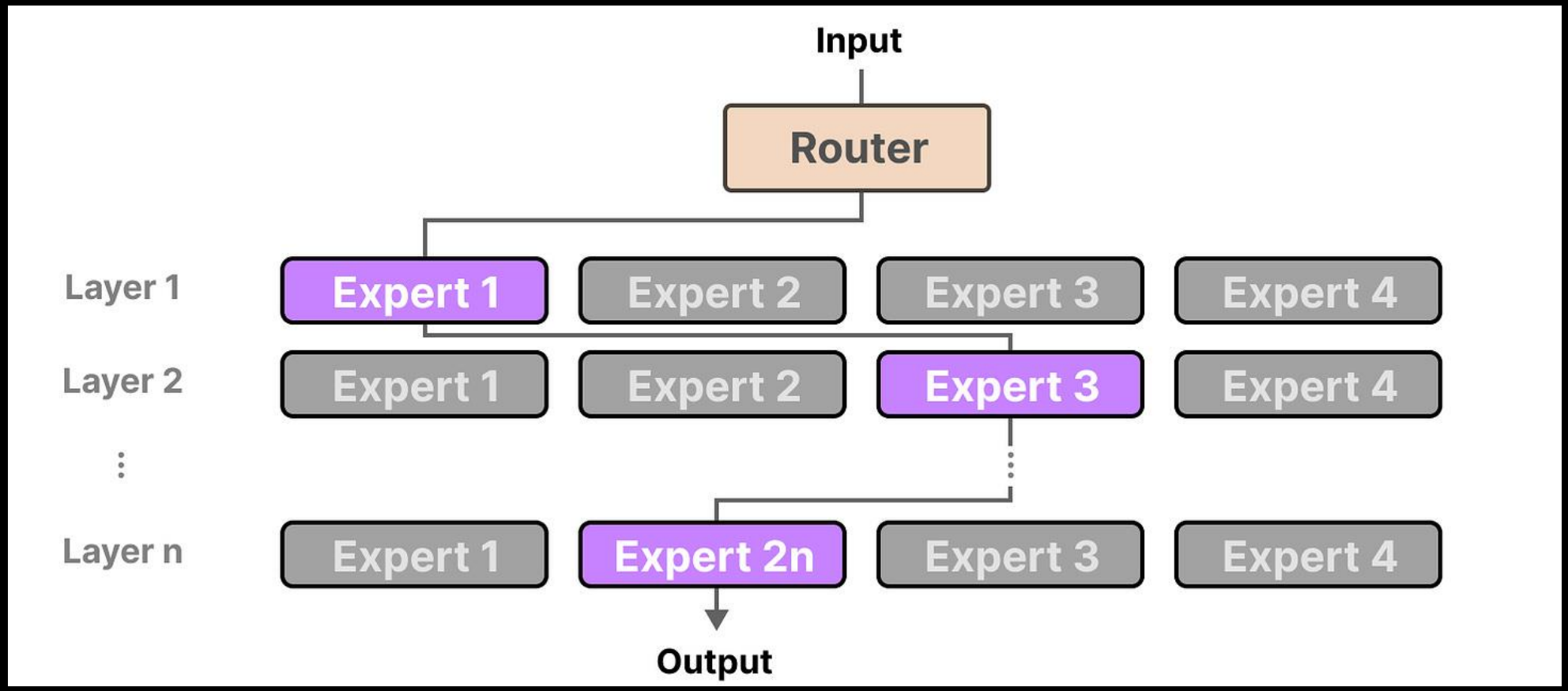
- Focus is on specific tokens, rather than topics (e.g., computer science)



https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mixture-of-experts

Mixture of Experts: Routers (aka, Gate Network)

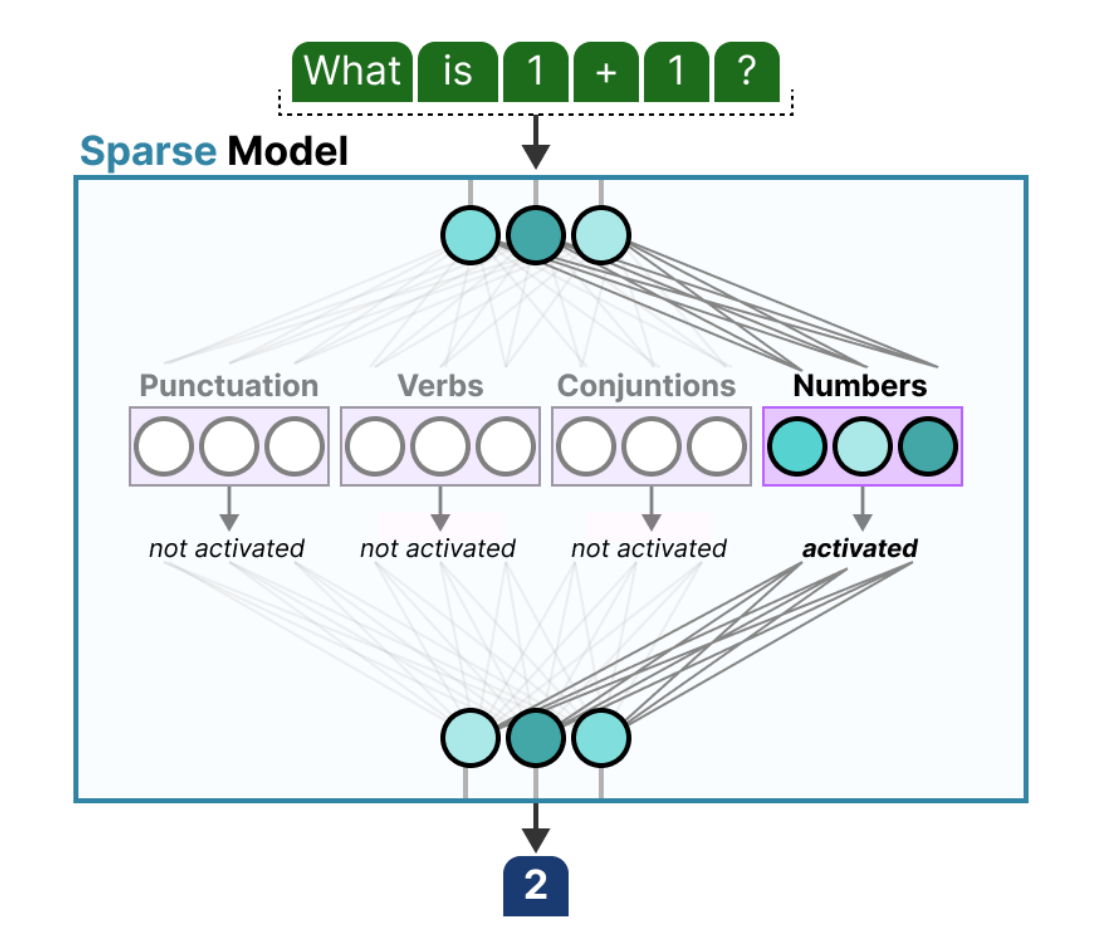
- Given an input, the router selects the best-suited expert(s)



https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mixture-of-experts

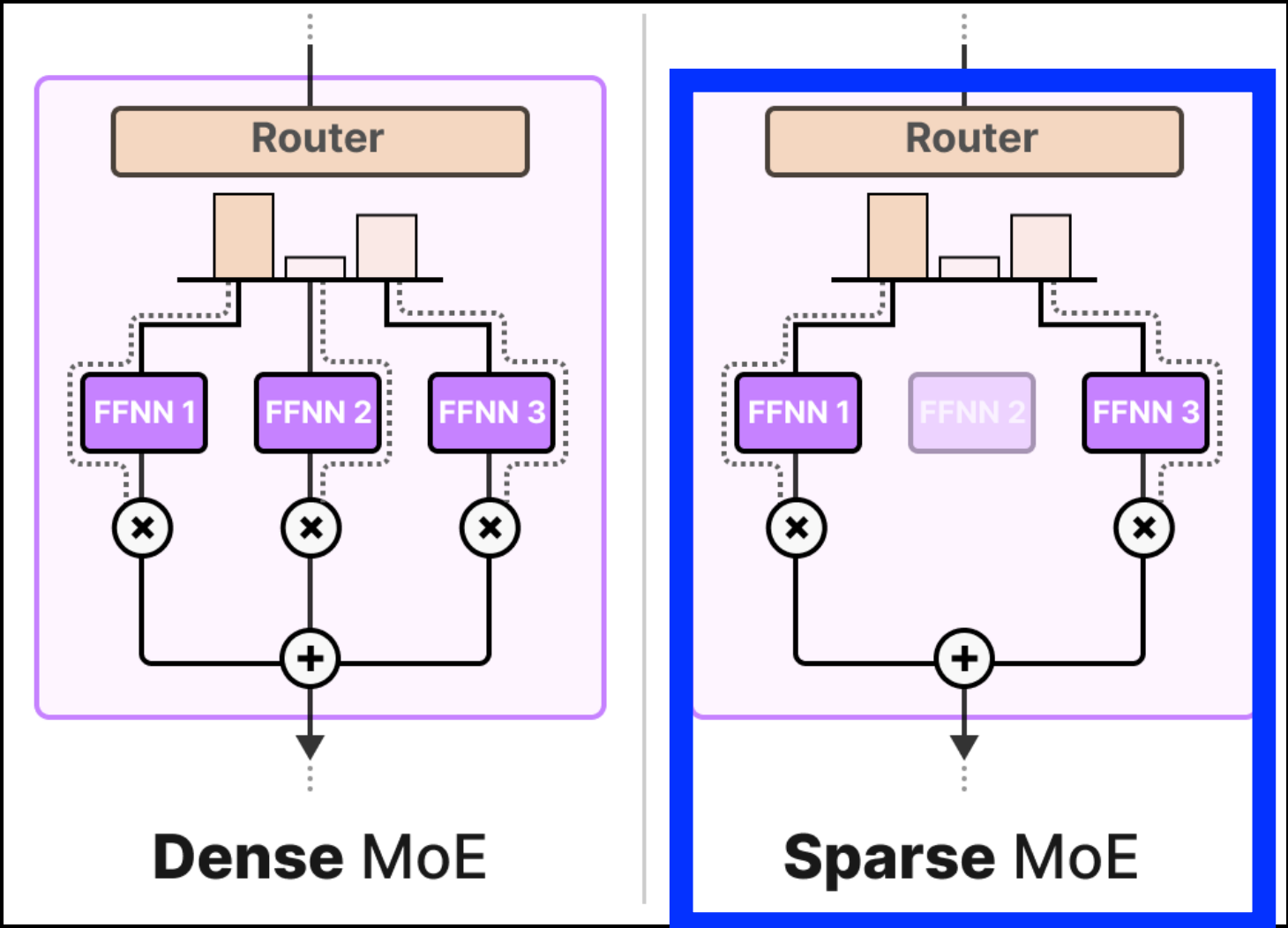
Mixture of Experts: Routers

- Given an input, the router selects the best-suited expert(s); e.g.,



https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mixture-of-experts

Mixture of Expert: Dense vs Sparse Layer

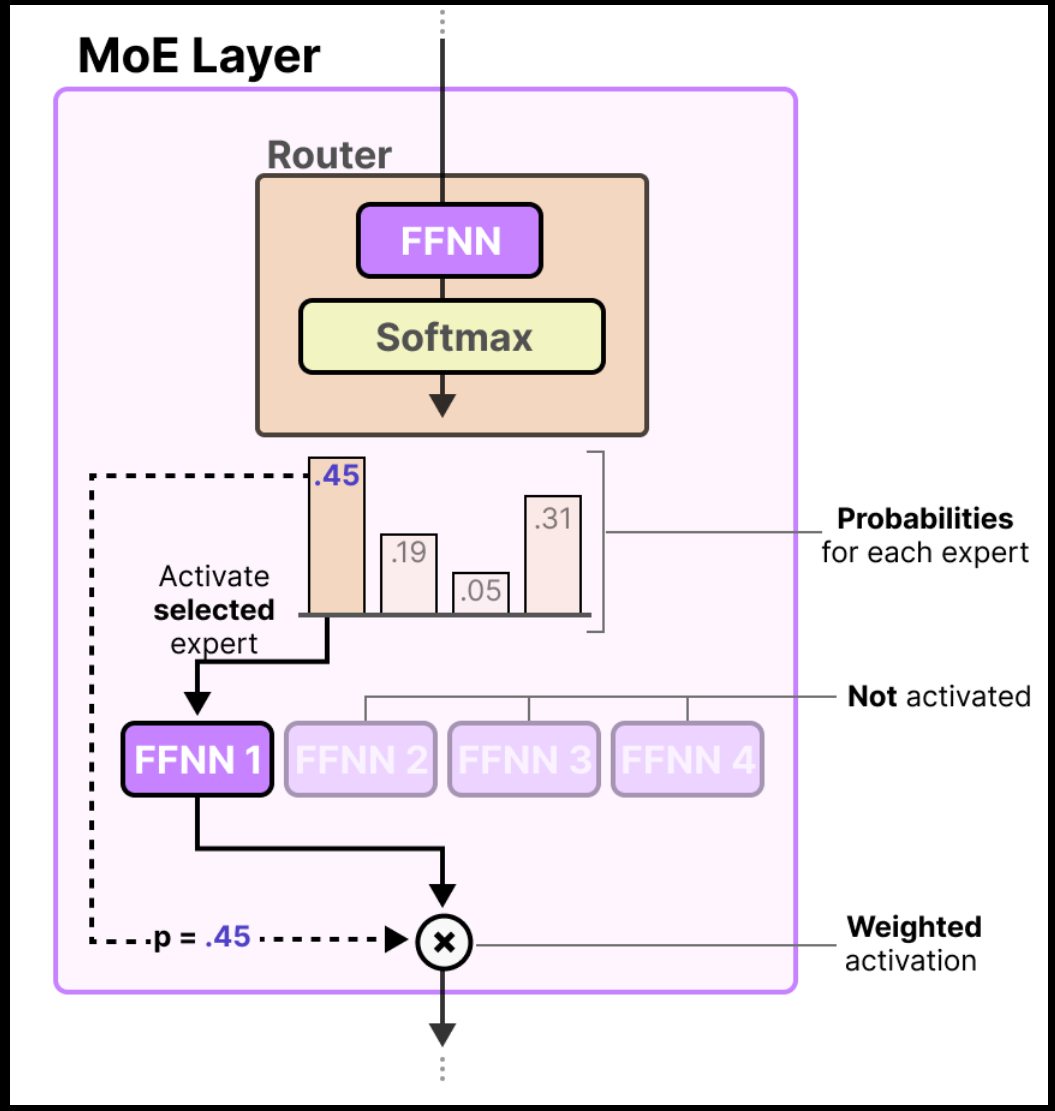


Beneficial for inference time!

Learning challenge: load balancing to apply similar importance to many experts rather than overfitting to few experts

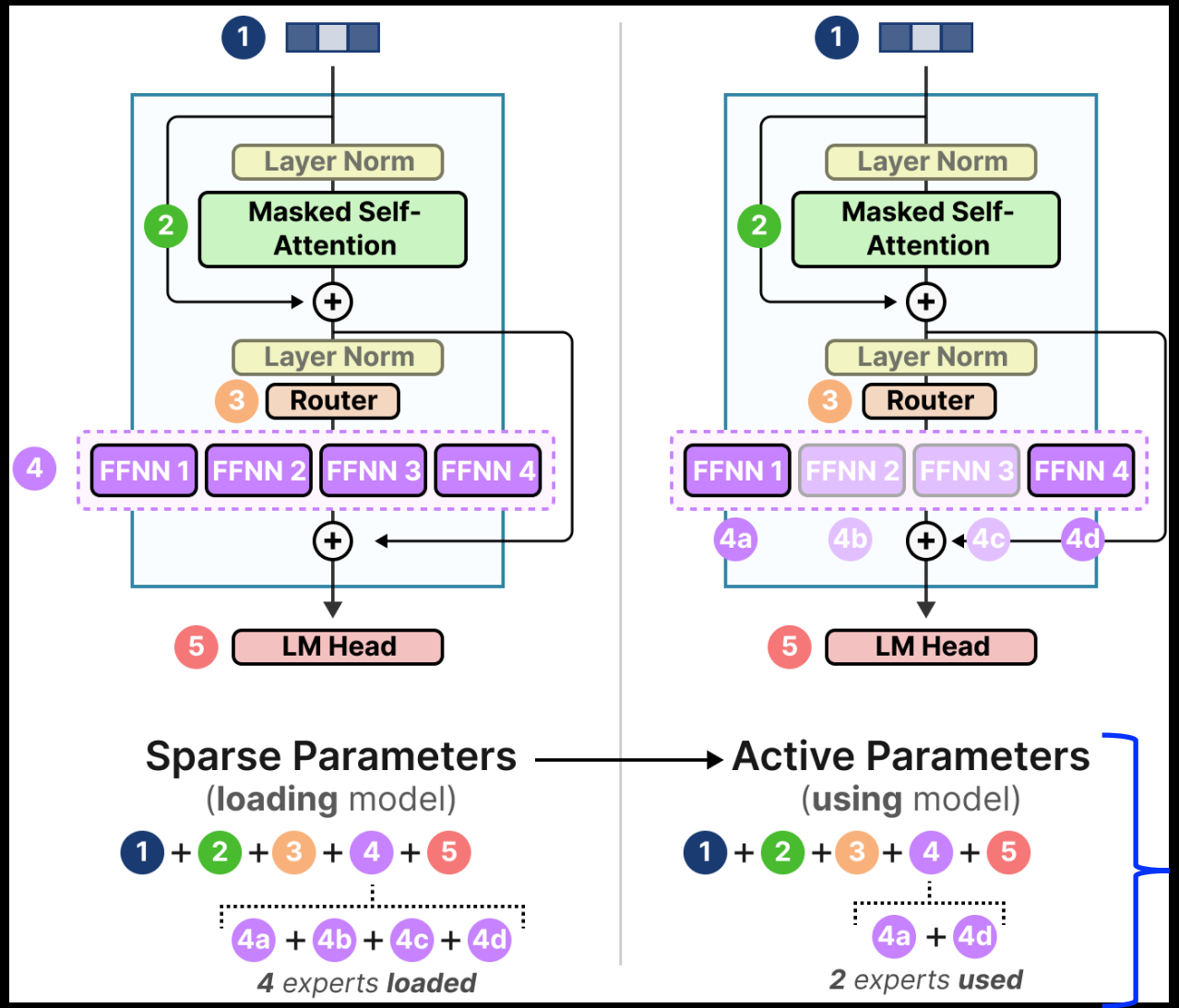
https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mixture-of-experts

Mixture of Experts: Router Implementation



https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mixture-of-experts

Mixture of Expert: Inference Time Set-Up

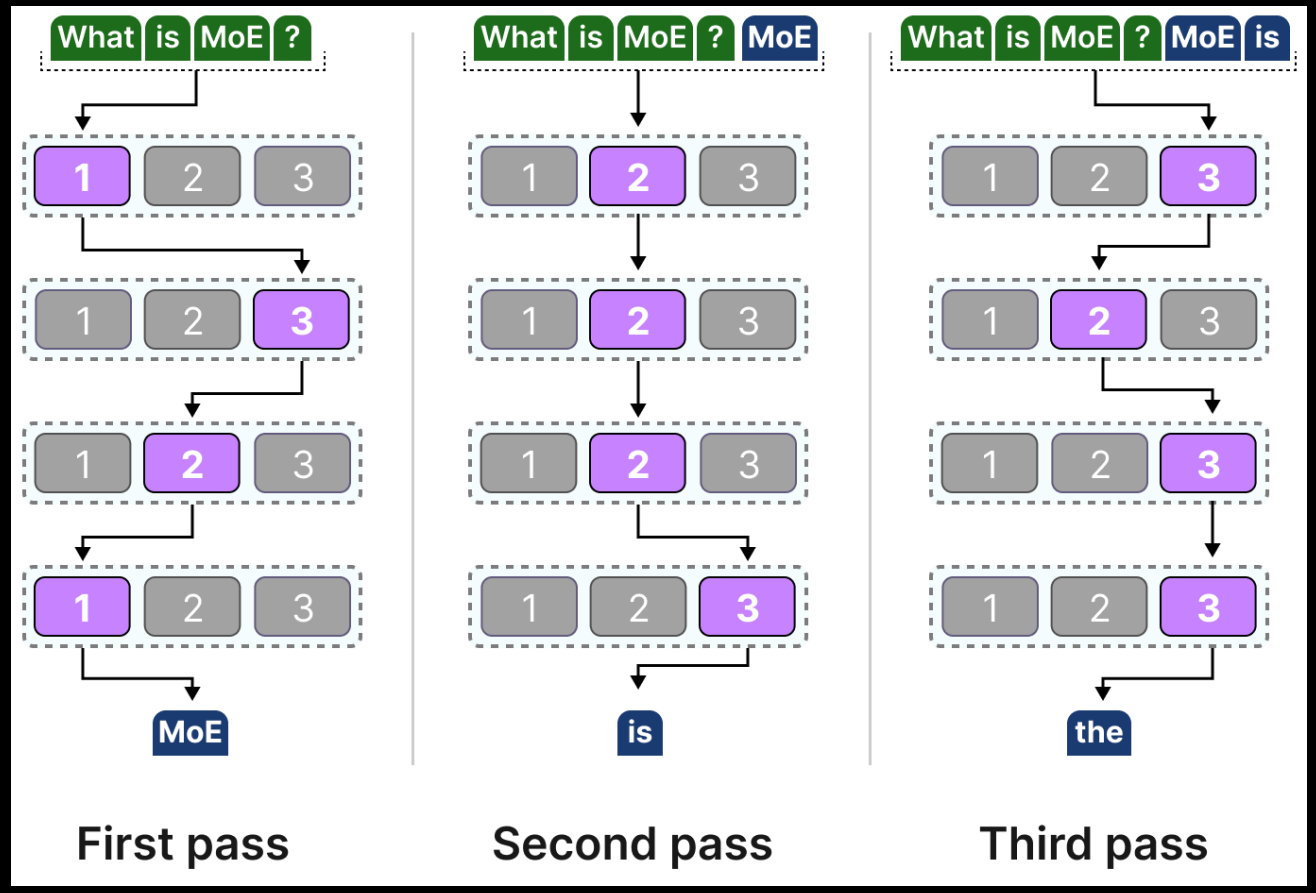


Fewer inference time computations!

https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mixture-of-experts

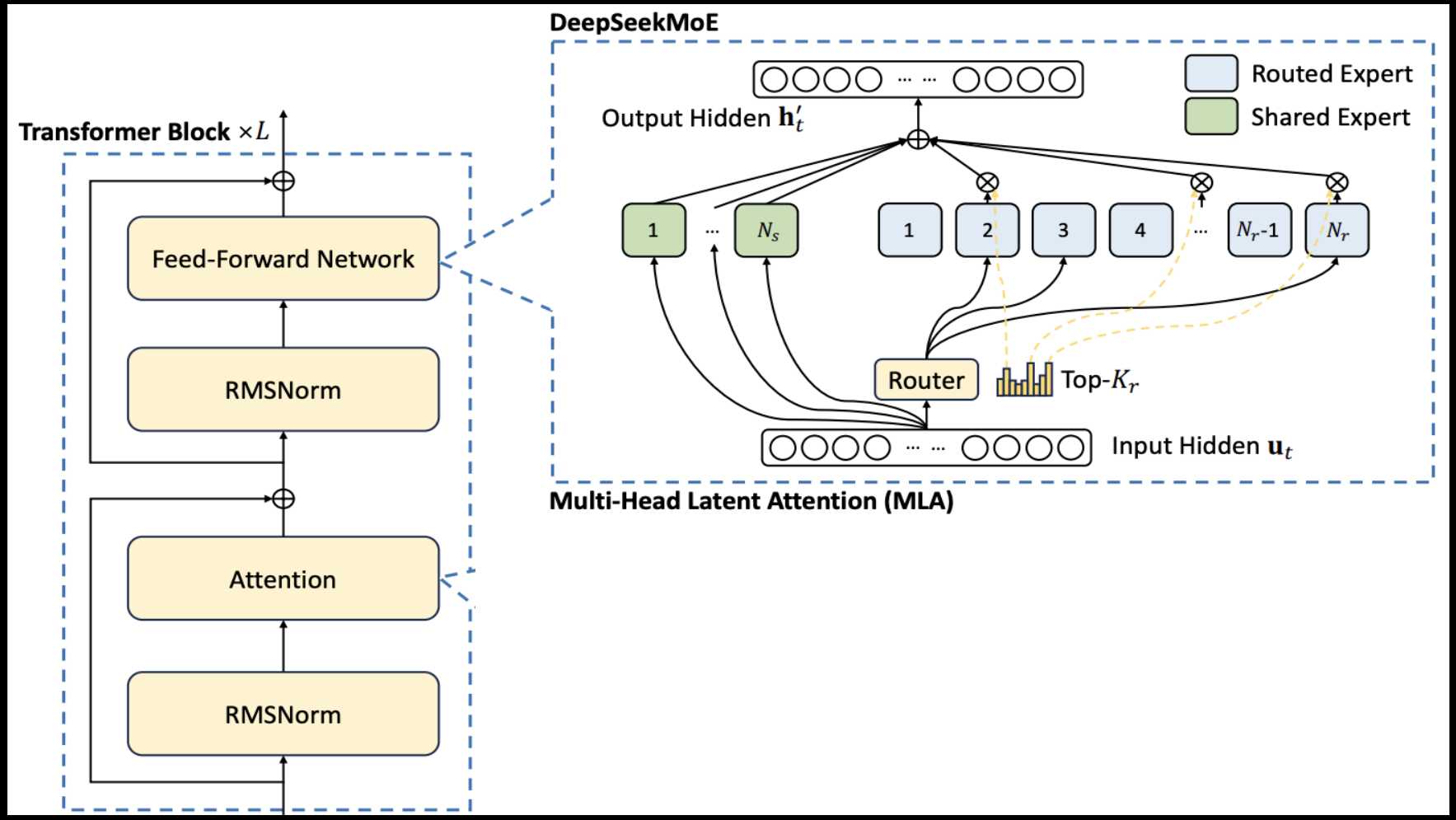
Mixture of Experts: Routers

- Paths can differ at different time steps for autoregressive models



https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mixture-of-experts

DeepSeek V3/R1 Implementation



From the 671B parameter model, activates ~37B parameters per token

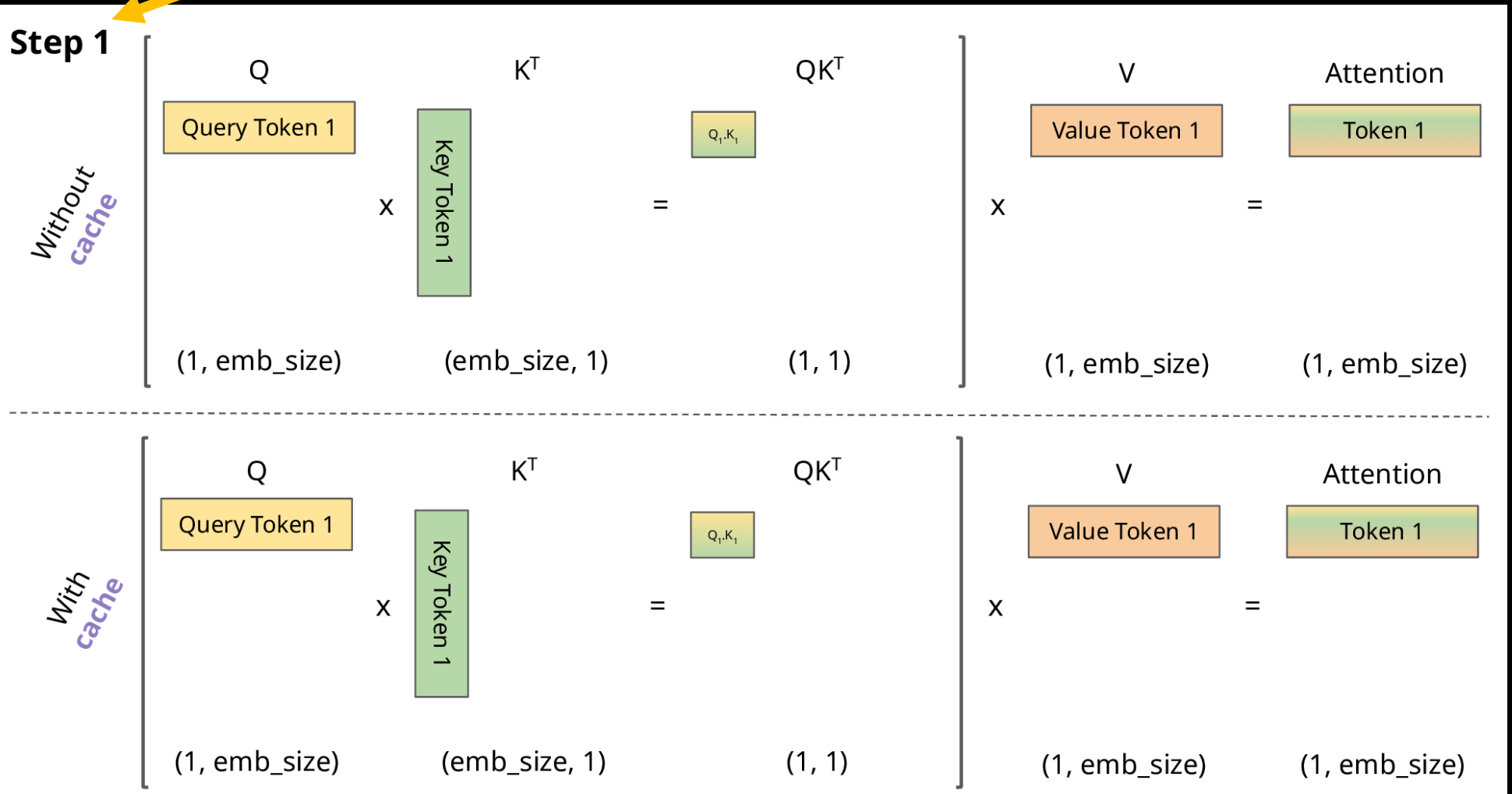
DeepSeek-V3 paper. 2024

Ideas

- Mixture of experts: reduce computation
- Multi-head latent attention: reduce computation
- Multi-token prediction: accelerate computation

Background: KV Caching (for Attention)

Only applies to generative (autoregressive) models



(Operations repeated in future time steps have results stored in cache)

Limitations: uses memory inefficiently since needed cache size regularly changes and reserves lots of cache memory to support large sizes

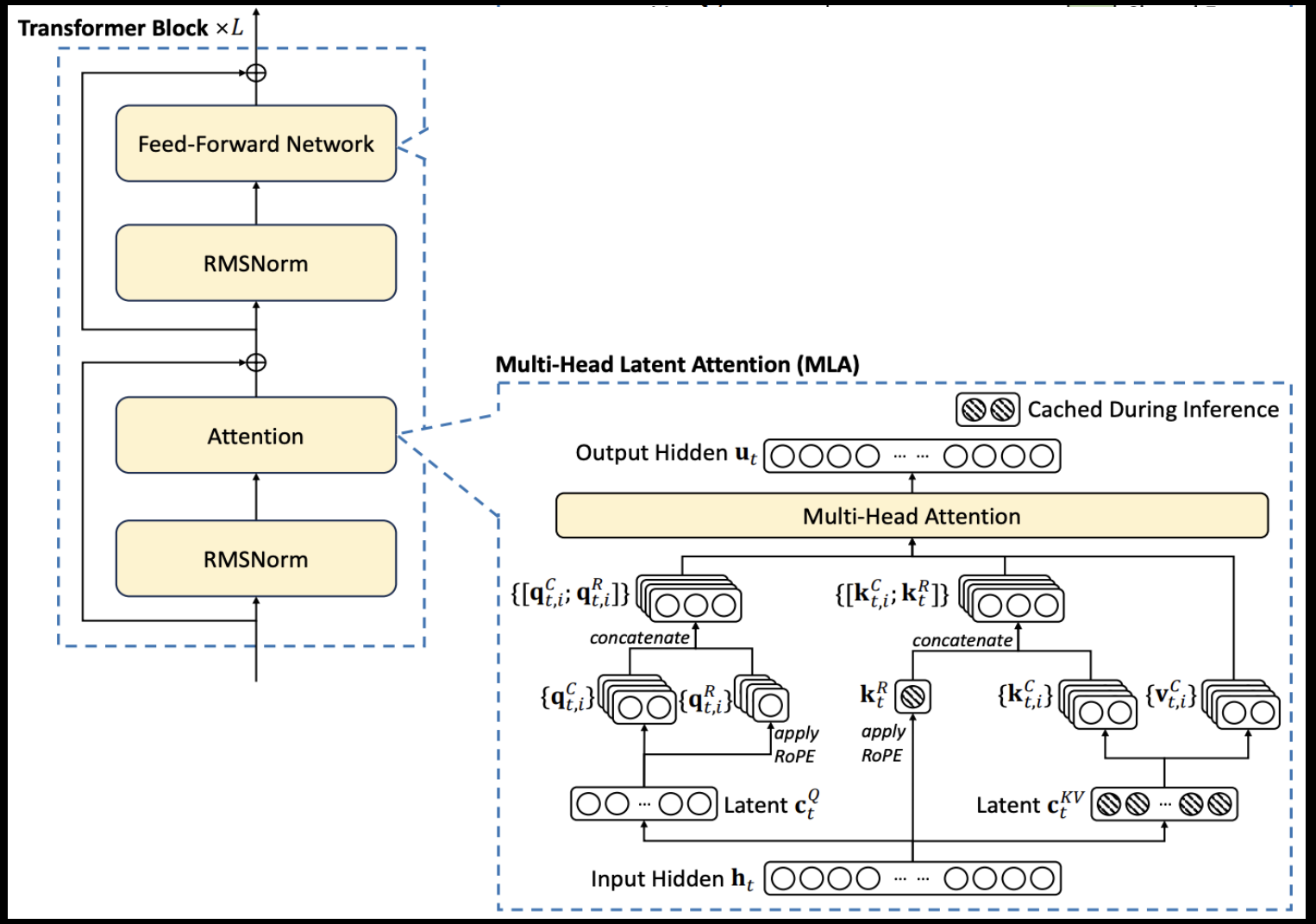
Store Keys and Values

Values that will be masked

Values that will be taken from cache

https://medium.com/@joaolages/kv-caching-explained-276520203249

V3/R1 Solution: Multi-Head Latent Attention



Architecturally, adds a small number of parameters to project KV cache to a smaller vector; only this vector must be cached at inference

DeepSeek-V3 paper. 2024

Also Relevant: vLLM Offers an Inference Trick

- Inspiration: virtual memory with paging used in operating systems to limit memory fragmentation by allocating memory “pages” on demand
- Idea: an LLM serving engine using PageAttention, which allocates KV cache into fixed-size blocks (aka, “pages”) that don’t have to be contiguous

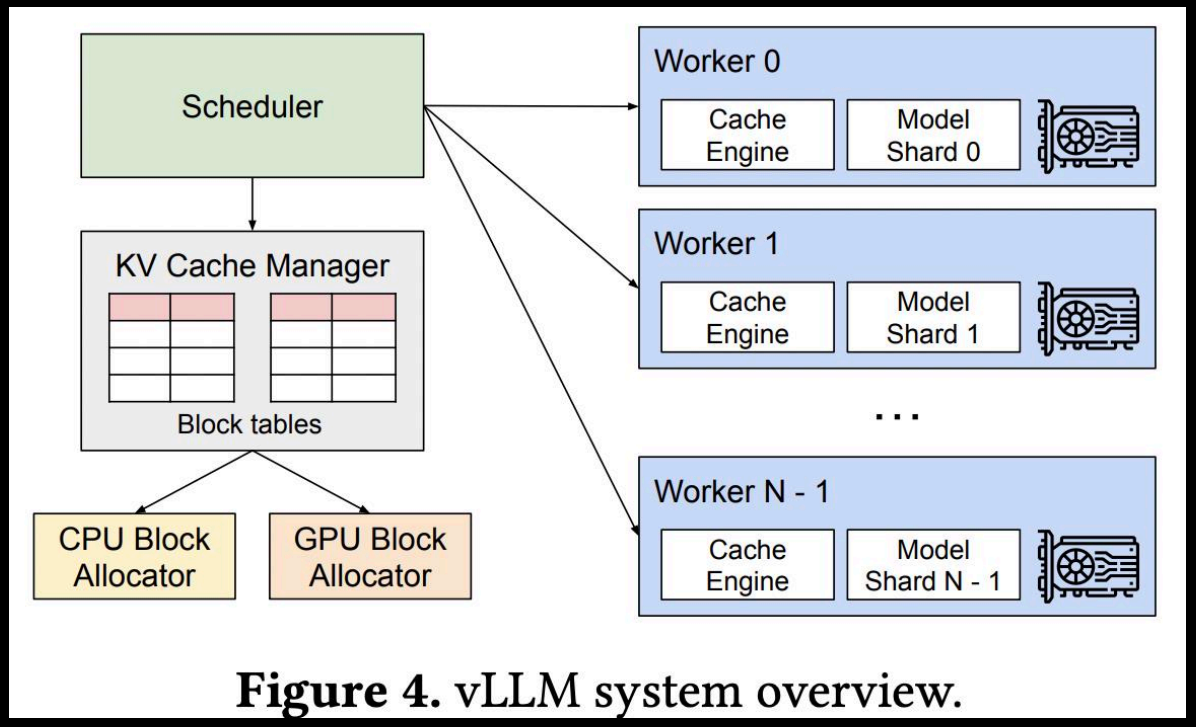


Figure 4. vLLM system overview.

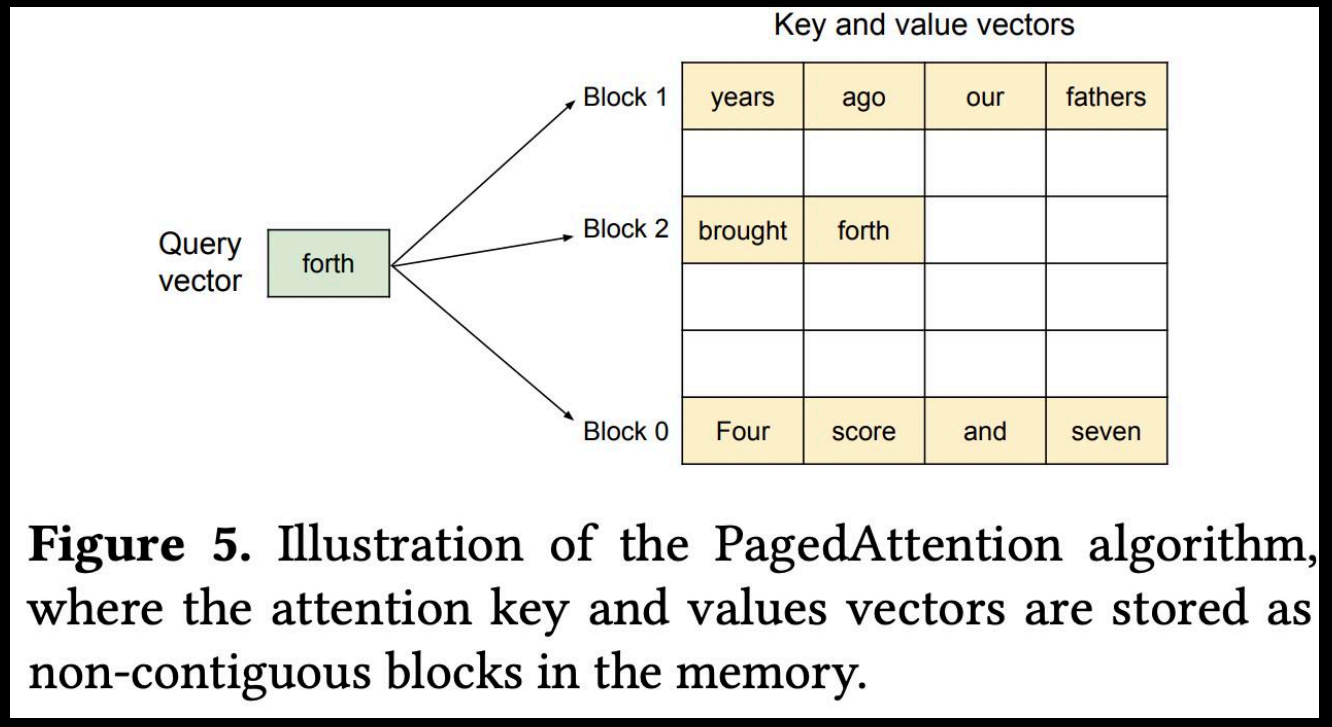


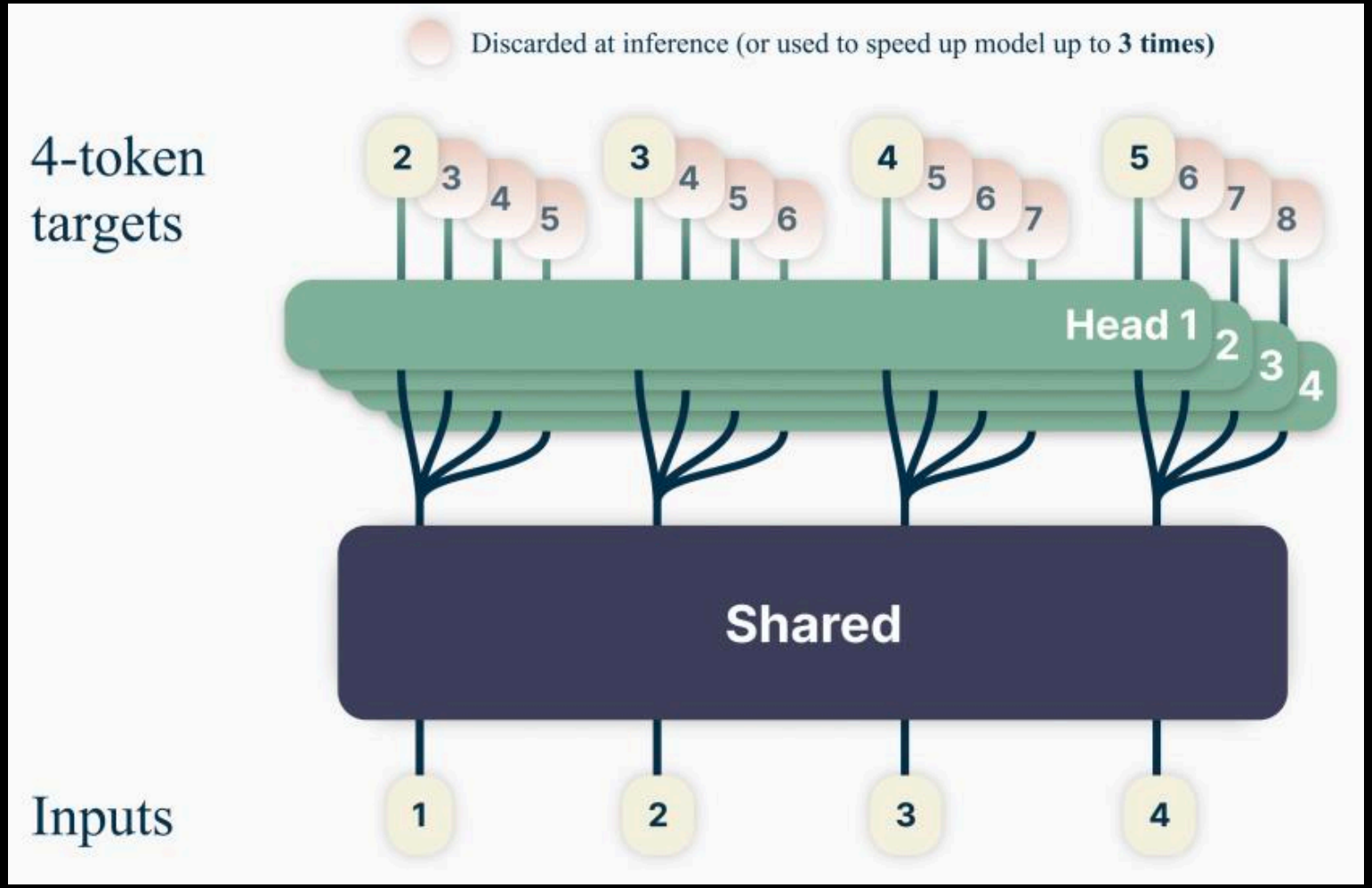
Figure 5. Illustration of the PagedAttention algorithm, where the attention key and values vectors are stored as non-contiguous blocks in the memory.

Kwon et al. Efficient Memory Management for Large Language Model Serving with PagedAttention. SOSP 2023

Ideas

- Mixture of experts: reduce computation
- Multi-head latent attention: reduce computation
- Multi-token prediction: accelerate computation

Multi-Token Prediction



Use *N* output heads to predict *N* tokens at each time step

Gloeckle et al. Better & Faster Large Language Models via Multi-token Prediction. April 2024

Ideas: Summary

- Mixture of experts: reduce computation
- Multi-head latent attention: reduce computation
- Multi-token prediction: accelerate computation

Today’s Topics

- The shock: frontier AI without the frontier budget
- Three technical levers for efficient intelligence
- The bigger story: a new scaling paradigm

A rapidly growing focus: efficient AI via hardware, architecture, and training.

Landscape currently feels like a sampler plate:



https://beyondmeresustenance.com/mexican-charcuterie-board/

Many Additional Options to Explore; e.g.,

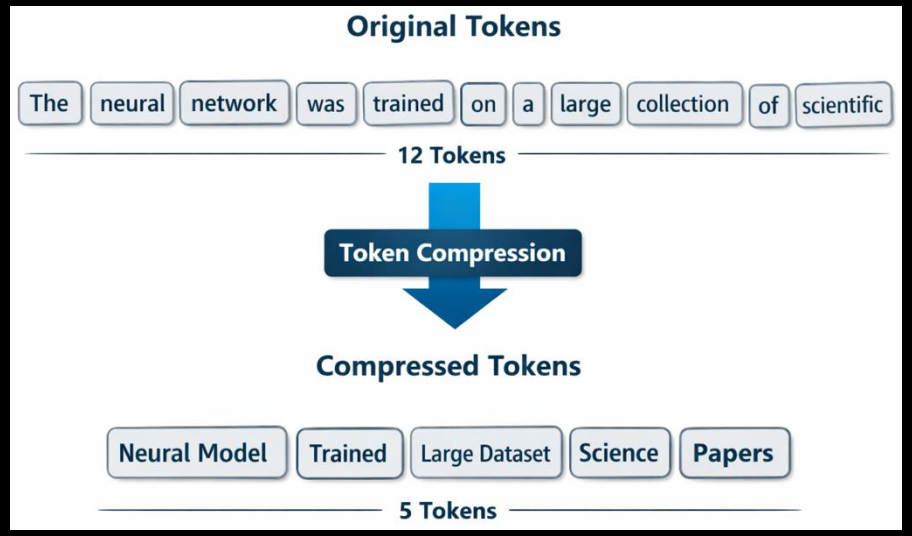
Token Filtering and Compression

Idea: verbosity is expensive

Saying 'Thank You' to ChatGPT Is Costly. But Maybe It's Worth the Price.

Adding words to our chatbot can apparently cost tens of millions of dollars. But some fear the cost of not saying please or thank you could be higher.

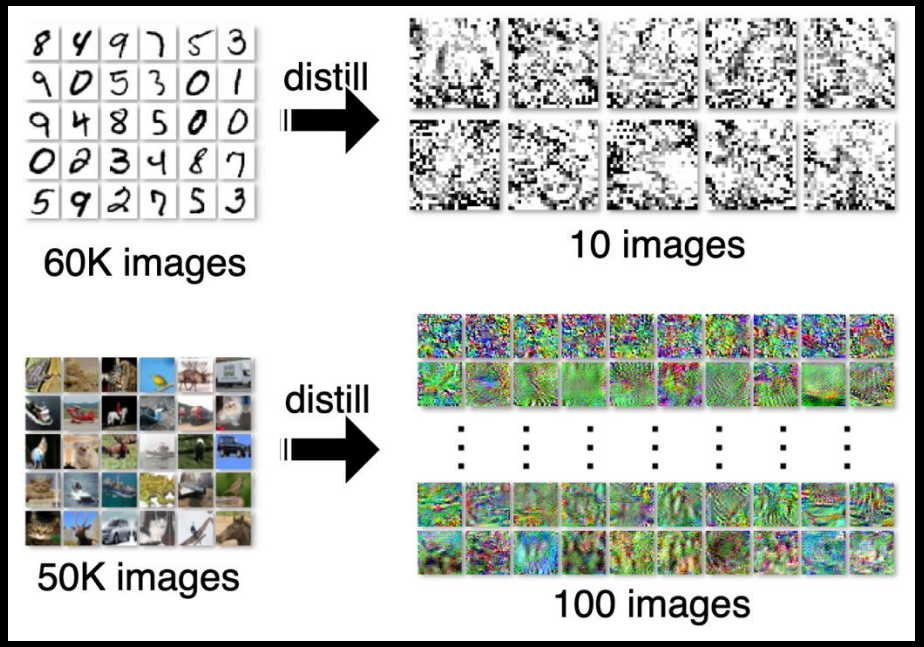
https://www.nytimes.com/2025/04/24/technology/chatgpt-alexa-please-thank-you.html



e.g., token pruning



Dataset Distillation



Approach: summarize dataset into a small number of examples that enable models to learn with less gradient updates and memory requirements

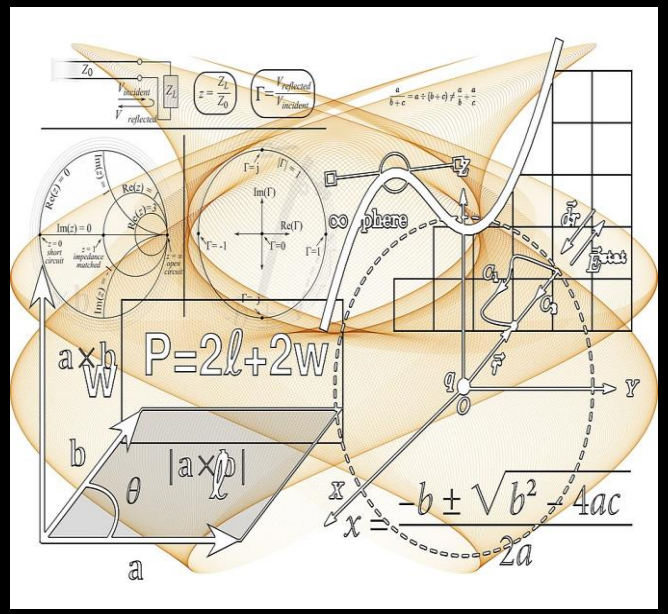
Wang et al. Dataset Distillation. arXiv 2018

Many Additional Options to Explore; e.g.,

Curriculum Learning

Intuition: how did you learn math?

Random Order of Examples



Ordered Examples



Big Book of Math; Dinah Zike

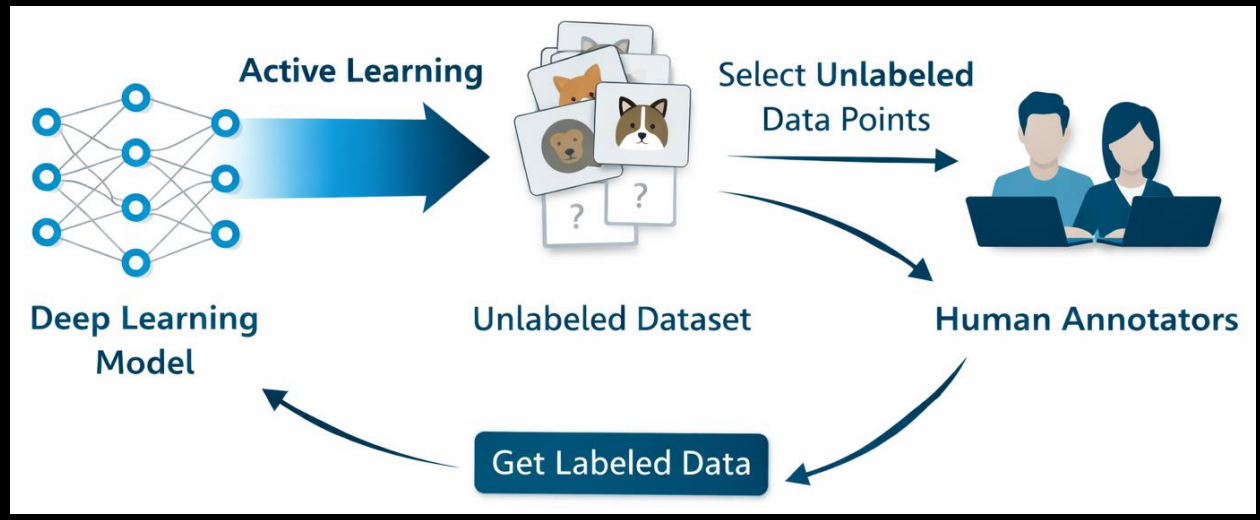
Approach: accelerate learning by presenting examples in order of increasing difficulty

Bengio et al., Curriculum Learning, 2009

Active Learning

Intuition: how do you effectively use your limited time with an expert?





Approach: choose which data to label based on which are anticipated to provide the most learning value

Efficient NN and DL: Not Only Better Sustainability But Also Enables New Applications

Learning and inference on-device for better privacy or resource-constrained settings (e.g., inadequate Internet connection)



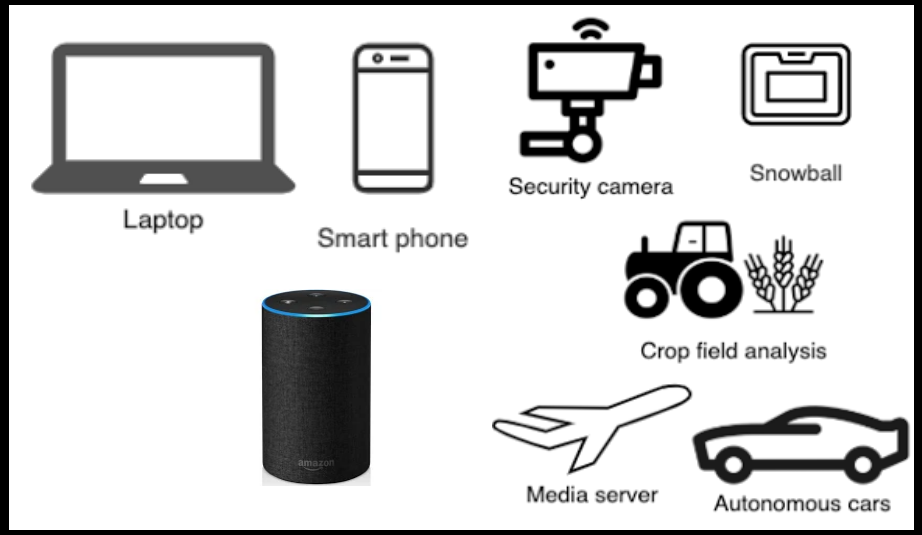
https://www.ephotozine.com/article/19-things-to-look-out-for-in-a-smartphone-camera--31055



https://en.wikipedia.org/wiki/Wearable_technology



https://www.buzzfeednews.com/article/katienotopoulos/facebook-is-making-camera-glasses-ha-ha-oh-no



https://aws.amazon.com/blogs/machine-learning/demystifying-machine-learning-at-the-edge-through-real-use-cases/

Today’s Topics

- The shock: frontier AI without the frontier budget
- Three technical levers for efficient intelligence
- The bigger story: a new scaling paradigm

The End